Electronic Cricket Scoreboard

Rishit Madireddy

SYSTEM PLANNING

PROBLEM STATEMENT

A problem that my cricket club faces is that the scoreboard at our home ground isn't electronically powered and requires the players on the bench to manually flip the scoreboard which is quite tedious. In previous matches, this meant that the scoreboard was initially used for the first few hours then remained untouched for the rest of the match, resulting in the incorrect score being displayed. My solution to this is an electronic scoreboard which will replace the existing scoreboard and shows the number of runs scored as well as the wickets lost by each team and can be controlled by the simple press of a button.

RESEARCH

During my research, I found that typical scoreboards displayed cricket scores in a 5 digit format where 1 of the digits was a dash to separate the number of runs and number of wickets. I also found out that for longer matches, the number of runs scored could go up to or sometimes even over 400. As a result, I decided to use five 7-segment displays where three display the number of runs, one separates the runs and wickets and one for the number of wickets. Since the scoreboard should be usable by either team, I decided to include a reset switch that can be used after one team's innings so that the score resets to 000-0 when pressed. Also, as longer matches can last beyond sunset when it might be more difficult to read the score from a distance, I decided to incorporate a row of LEDs that can be arranged around the scoreboard so that the score remains visible even after sunset. From what I read, I found out that typical light intensity at sunset is around 400 lux but I wanted the LEDs to turn on a few minutes prior to sunset so I decided that the row of LEDs should light up when light intensity falls below 425 lux.

SPECIFICATION

Quantitative:

1. The scoreboard should have at most 5 buttons (1 run, 4 runs, 6 runs, 1 wicket and reset) to remain relatively simple to operate.
2. The scoreboard should consist of 5 7-segment displays (3 for runs, 1 for a dash and 1 for wickets)
3. The current through each segment should be 4mA ± 0.5mA so that the score displayed is clearly visible
4. The power dissipated by each white LED should be 5mW ± 0.5mW.
5. The row of LEDs should turn on when the light level drops to below 425 ± 10 lux.
6. The current through the red LED should be 7mA ± 0.5mA.
7. The frequency of the astable should be close to 1Hz ± 0.5Hz.
8. The duty cycle of the astable should be close to 50% ± 5%.

Qualitative:

1. All resistors used must be in the E24 series.
2. There should be 2 scores visible - one for runs and one for wickets.
3. The runs scoreboard should be able to allocate 3-digit scores (i.e. up to 999).
4. The scoreboard should run primarily with 555 ICs in either the astable or monostable configurations.
5. The 1 run button should add 1 to the runs score when pressed.
6. The 4 runs button should add 4 to the runs score when pressed.
7. The 6 runs button should add 6 to the runs score when pressed.
8. The 1 wicket button should add 1 to the wickets score when pressed.
9. The reset button should reset both scores so that they are both zero.
10. There should be a row of white LEDs which turn on if it gets dark so that the score is still visible.
11. Each time a wicket is lost, a red LED should light up.
12. The scoreboard should be able to run off of a 9V battery but can be tested with 5V d.c.

ANALYSIS OF PROBLEM

Based on my problem statement and the specification, I decided to base my design off of 555 ICs. In particular, I decided to use an astable and 4 monostables - 3 for scoring 1 run, 4 runs and 6 runs and 1 for scoring 1 wicket. As for how this would work, my plan was to AND the outputs of the monostable and astable so that when the 4 runs button was pressed, for example, the time period of the output of the monostable should last 4 rising edges of the monostable so each time the astable went high, it caused the value displayed on the 7-segment display to increment. I applied this to all 4 monostables, making sure that the time period was appropriate for the number of runs/wickets that the score should increase by. The red LED is connected to the output of the 1 wicket monostable so that it not only lights up when a wicket is lost but also stays on for the same time period as the monostable so that it’s a bit more prominent. As for the decoder/driver for the 7-segment, I decided to use a 4026 as this IC included a decade counter so the divide by 10 output can be used as the clock input for the other 4026s. Lastly, I decided to use a light sensing subsystem consisting of an LDR and op-amp connected to the row of white LEDs so that the analogue input of the LDR would be converted to a binary output for the row of white LEDs.

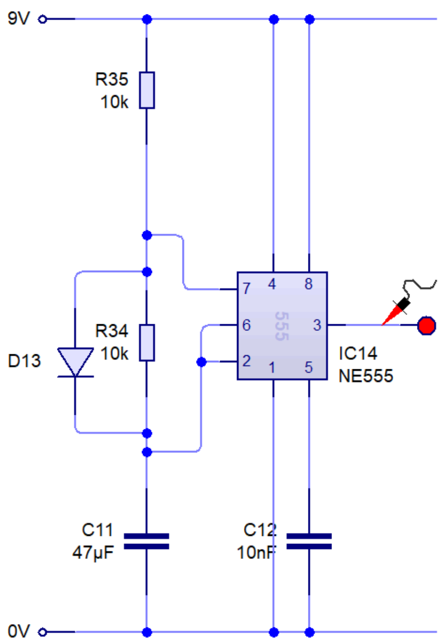
SYSTEM DEVELOPMENT

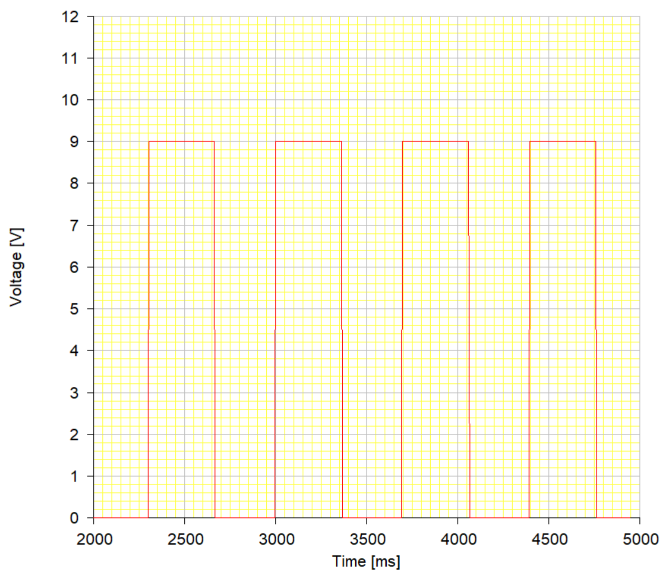
SUBSYSTEM 1 - 1HZ ASTABLE (DIGITAL)

Specification

As per the specification, the scoreboard should run primarily based on 555 ICs so I decided that the backbone of my scoreboard would be an astable that is constantly oscillating at a frequency of 1Hz ± 0.5Hz. To calculate the time periods of the monostables easily, the duty cycle of the astable should be close to 0.5 ± 0.05. All resistors used should also be part of the E24 series.

Subsystem circuit diagram



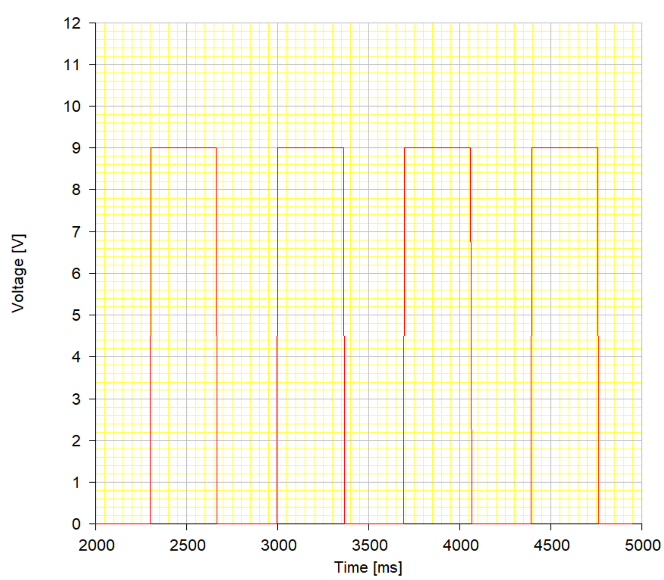


Testing

Test conditions	Expected output	Actual output value	Pass/Fail
R_1 = 10kΩ R_2 = 10kΩ C = 47μF	Frequency of 1Hz ± 0.5Hz and duty cycle of 50% ± 5%	Frequency of 1.42Hz with duty cycle of 51.9%	Pass

Report on performance

As shown by the results, the frequency of the astable is 1.42Hz. The original frequency of the astable can be calculated using the formula: $f = 1.44/C(R_1 + 2R_2)$ where the values of R_1 and R_2 and C in my circuit are 10kΩ, 10 and 47μF respectively. This gives a frequency of 1.02Hz. However, the time high is 0.651s and the time low is 0.326s and when I substituted these values into the equation T_H/T_H+T_L to calculate the duty cycle, this gave me 0.667 which is too high. Therefore, I decided to use a diode placed parallel with R_2 to give a new frequency of 1.42Hz, a time high of 0.366s, a time low of 0.339s and thus a duty cycle of 0.519 which is acceptable. The use of the diode means that R_2 won't be taken into account when the capacitor charges up but will when the capacitor discharges - fixing the issue with the duty cycle. The graph above shows the output of the astable when the circuit is initially powered, with values being representative of what I calculated. It also shows a longer time high upon startup as the capacitor is initially at 0V and hence takes one cycle to adjust. However, the time high is corrected soon after for the upcoming cycles (as shown below).

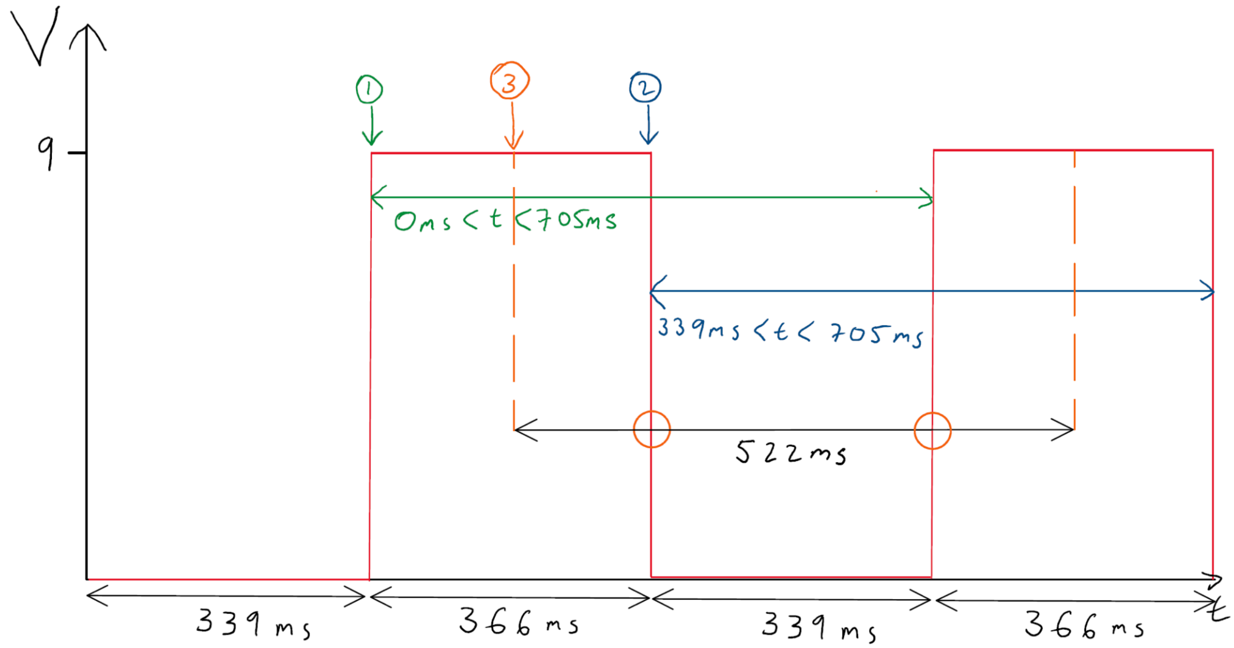


SUBSYSTEMS 2, 16, 19 - 1s MONOSTABLE (DIGITAL) + RED WICKET LED

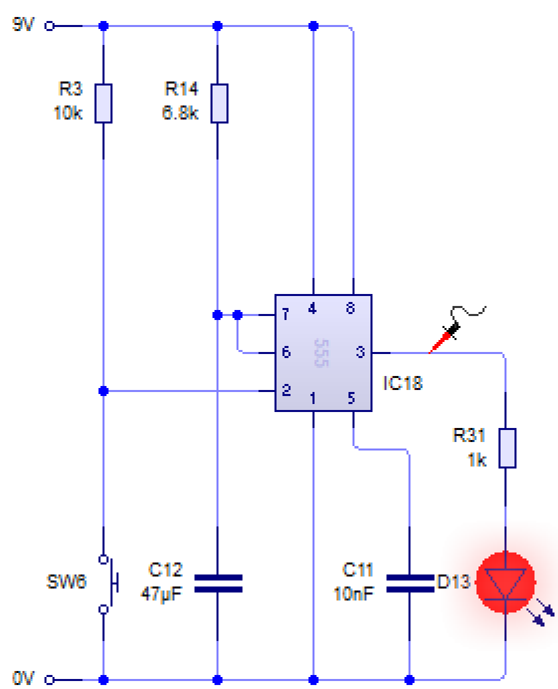
Specification

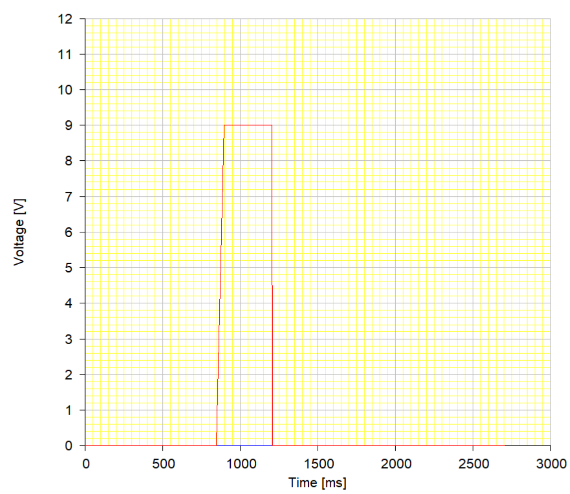
As per the specification, the scoreboard should run primarily based on 555 ICs so I wired 2 of them up in the monostable configuration and are responsible for adding 1 run and 1 wicket to the scoreboard. Also, the scoreboard should have 5 buttons at most and the switches on the trigger pin of the 1s monostables make up two of them. The output of the monostable responsible for 1 wicket should also make a red LED when triggered with the current through it being 7mA ± 0.5mA. All resistors used should also be part of the E24 series.

The timing in particular should be quite precise as it should be long enough to cover just one rising edge of the astable as any longer could cause it to cover just over 2 rising edges, causing the score to increment by 2 instead of 1. To find out the exact time period required, I decided to split the output of the astable into 2 “extreme” scenarios. Scenario 1 is where the monostable is triggered at the start of the time high of the astable (also the end of the time low) and scenario 2 is when the monostable is triggered at the end of the time high of the astable (also the start of the time low). In the case of scenario 1, the time period of the monostable should last from 0ms up to 705ms. This is because, as the astable is already high, a pulse of any duration from the monostable would cause the score to increment by 1, giving a lower bound of 0ms. When the time high passes, the time period should last until the next rising edge which covers the full time period of the astable, giving an upper bound of 705ms. In the case of scenario 2, the time period of the monostable should last from 339ms up to 705ms. This is because, as the astable is low, the monostable time period should be long enough until the rising edge which is essentially the time low of the monostable, giving a lower bound of 339ms. When the time low passes, the time period can last up to the end of the time high of the astable which, again, gives an upper bound of 705ms. After combining both inequalities, I found that the time period of the monostable should be between 339 to 705ms. So initially, I decided to test with a time period halfway between 339 and 705ms, 522ms. And after using the correct RC values and thorough testing, I found that I didn’t account for another “extreme” scenario where the monostable could be triggered halfway through the time high period of the astable. When I tested this scenario, I found that the score rose by 2 instead of 1 as the first rising edge of the monostable initially incremented the score by 1 and the second rising edge occurred because of the astable once it had gone low and was going high for the next cycle. Therefore, I decided that the time period should be close to 339ms ± 15ms - giving a very small chance of error. I have provided a diagram below to illustrate the above, with the scenarios labelled based on their number.



Subsystem circuit diagram





Testing

Test conditions	Expected output	Actual output value	Pass/Fail
R_1 = 6.8kΩ C = 47μF	Time period of 339ms ± 15ms	Time period of 351.56ms	Pass
R_1 = 6.8kΩ C = 47μF	Red LED should light up when triggered with current of 7mA ± 0.5mA	Red LED lights up when triggered with current of 6.89mA	Pass

Report on performance

The results show that the time period of the monostable is 351.56ms which is just over 339ms by 12.56ms and is hence an acceptable time period. Although, this does mean that if the monostable is triggered coincidentally when the astable is in the last 12.56ms of its time high period, the score could increase by 2. However, 12.56ms is very small so the chance of this happening is very small and is supported by my testing where the score only incremented by 1 in all of my test trials for the monostable. Also, we can see that the red LED lights up when the 1 wicket monostable is triggered. To add, I decided to use a 1kΩ load resistor to protect the LED from excessive current.

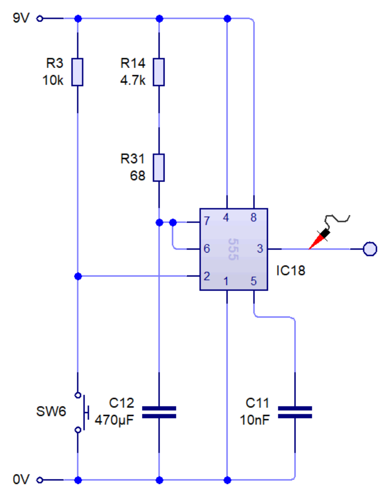
SUBSYSTEM 3 - 4s MONOSTABLE (DIGITAL)

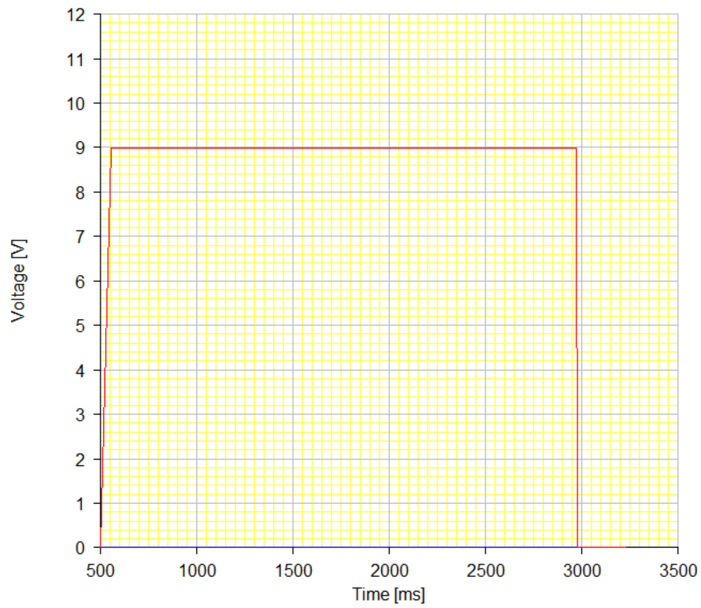
Specification

As per the specification, the scoreboard should run primarily based on 555 ICs so I wired the 4 runs 555 in the monostable configuration and is responsible for adding 4 runs to the scoreboard. Also, the scoreboard should have 5 buttons at most and the switch on the trigger pin of the monostable makes up one of them. All resistors used should also be part of the E24 series.

As for the timing, I used the same thought process for the 1 run monostable and decided that the time period can be between 2.454s and 3.159s. This time, taking the third scenario into account, the time period should therefore be 2.454s ± 0.015s.

Subsystem circuit diagram





Testing

Test conditions	Expected output	Actual output value	Pass/Fail
R_1 = 4.7kΩ+68Ω = 4.768kΩ C = 470µF	Time period of 2.454s ± 0.015s	Time period of 2.465s	Pass

Report on performance

The results show that the time period of the monostable is 2.465056s which is just over 2.454s by 11.06ms and is hence an acceptable time period. Although, this does mean that if the monostable is triggered coincidentally when the astable is in the last 11.06ms of its time high period for its 4th cycle, the score could increase by 5. However, 11.06ms is very small so the chance of this happening is very small and is supported by my testing where the score only incremented by 4 in all of my test trials for the monostable.

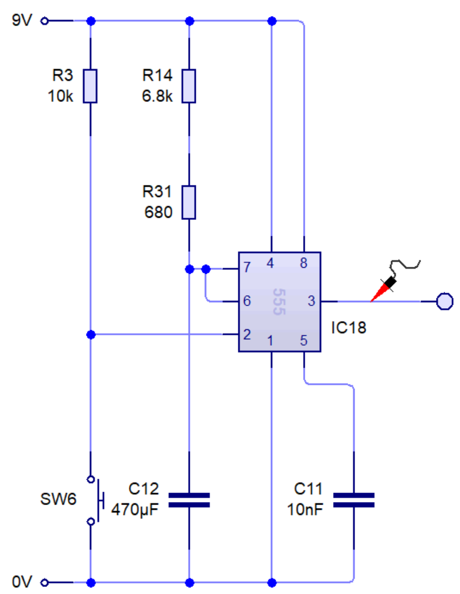
SUBSYSTEMS 4 - 6 runs MONOSTABLE (DIGITAL)

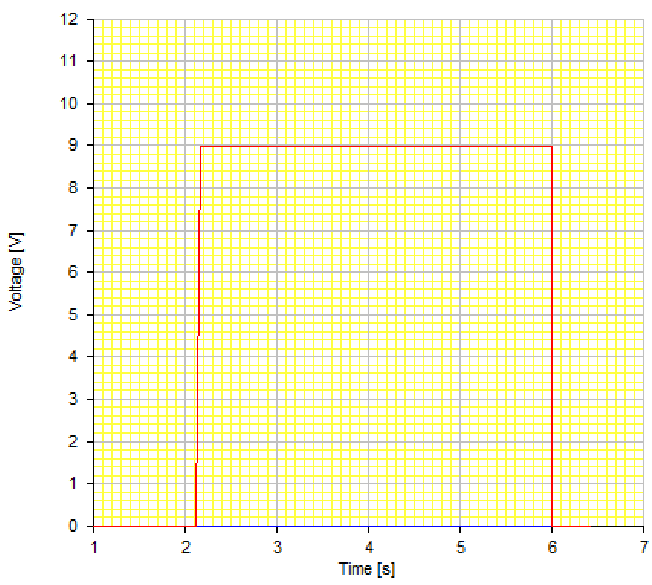
Specification

As per the specification, the scoreboard should run primarily based on 555 ICs so I wired the 6 runs 555 in the monostable configuration and is responsible for adding 6 runs to the scoreboard. Also, the scoreboard should have 5 buttons at most and the switch on the trigger pin of the monostable makes up one of them. All resistors used should also be part of the E24 series.

As for the timing, I used the same thought process for the 1 run monostable and decided that the time period can be between 3.864s and 4.23s. This time, taking the third scenario into account, the time period should therefore be 3.864s ± 0.015s.

Subsystem circuit diagram





Testing

Test conditions	Expected output	Actual output value	Pass/Fail
R_1 = 6.8kΩ+680Ω = 7.48kΩ C = 470µF	Time period of 3.864s ± 0.015s	Time period of 3.867s	Pass

Report on performance

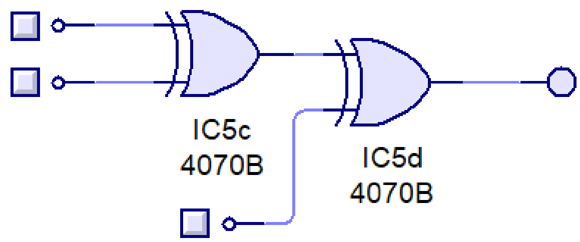
The results show that the time period of the monostable is 3.867s which is just over 3.864s by 3.16ms and is hence an acceptable time period. Although, this does mean that if the monostable is triggered coincidentally when the astable is in the last 3.16ms of its time high period for its 6th cycle, the score could increase by 7. However, 3.16ms is very small so the chance of this happening is very small and is supported by my testing where the score only incremented by 6 in all of my test trials for the monostable.

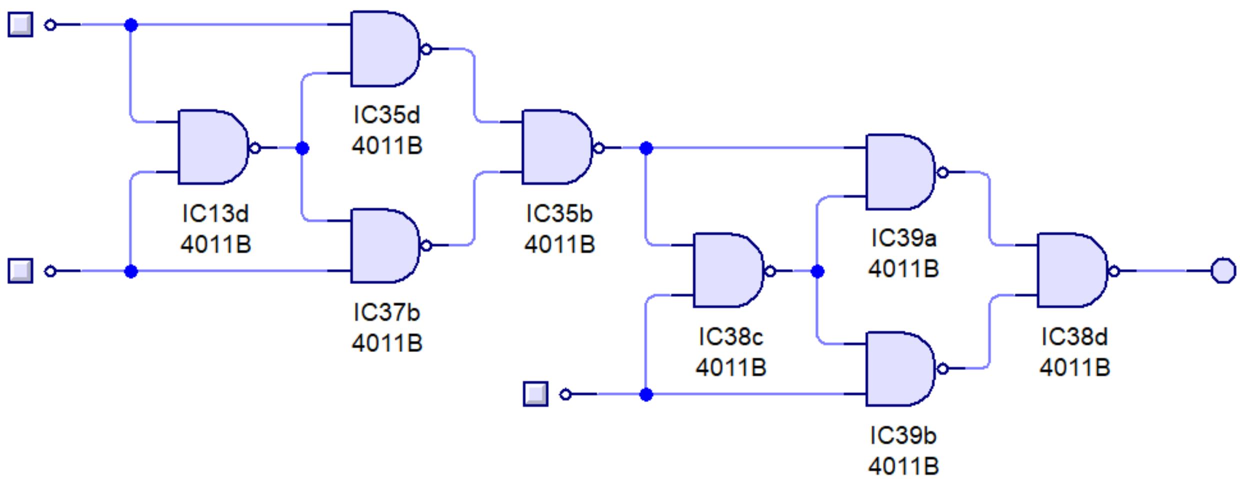
SUBSYSTEM 5 - 3 INPUT XOR GATE (DIGITAL)

Specification

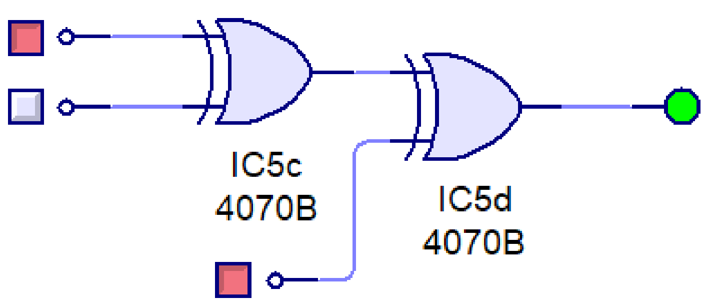
To combine the outputs of the various monostables such that only one can be triggered at a time and nothing happens when 2 or more are triggered, I decided to use a 3 input XOR gate. To make the 3 input XOR gate, I cascaded two, 2 input XOR gates with the output of one connecting to the input of the next. I then converted these gates into NAND gates which is what I used when I physically prototyped the scoreboard on a breadboard. But for the purpose of testing, I will be using the original layout with the cascaded XOR gates.* I expect the XOR gate to provide a logic level of 0 when two or more inputs are HIGH except for the case where all three inputs are HIGH in which case I would expect a logic level of 1. I also expect the output to be LOW when all inputs are LOW and HIGH when only one input is HIGH.

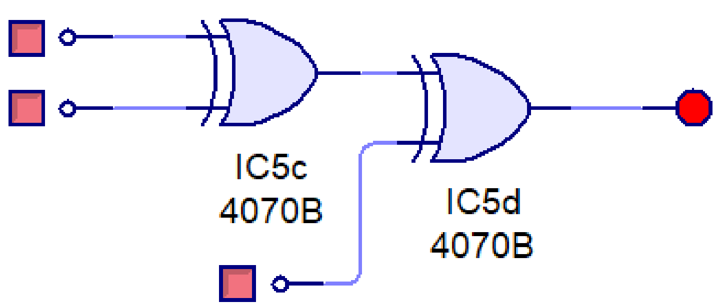
*The NAND gate conversion can be seen in the diagram below.





Subsystem circuit diagram





Testing

Test conditions	Input 1 run	Input 4 runs	Input 6 runs	Expected output	Actual output	Pass/Fail
All monostables low	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
Only 6 runs monostable high	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Pass
Only 4 runs monostable high	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 1 (9V)	Pass
Only 4 and 6 runs monostables high	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
Only 1 run monostable high	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 1 (9V)	Pass
Only 1 and 6 runs monostables high	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
Only 1 and 4 runs monostables high	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
All monostables high	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Pass

Report on performance

As shown by the table above, the results are consistent with my expected outcomes stated in my specification where the output is only HIGH when only 1 monostable's output is HIGH, except for the rare instance where all three monostables are triggered and the output of the XOR is HIGH. However, since this is even less likely than triggering 2 monostables by accident, this outcome isn't problematic to how the scoreboard works. Also, we can see that the table above matches with the truth table for a typical 3-input XOR gate.

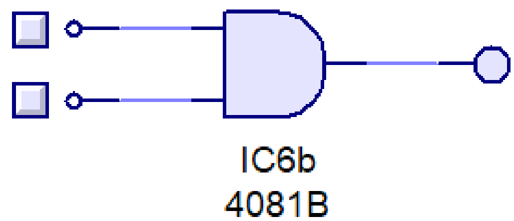
Inputs			Output
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

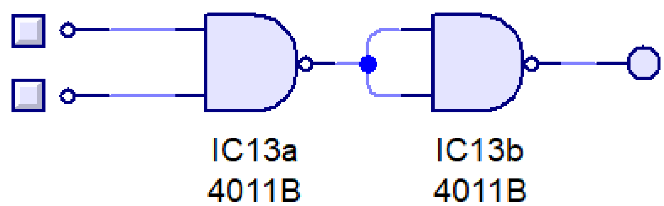
SUBSYSTEM 6 - AND GATE (DIGITAL)

Specification

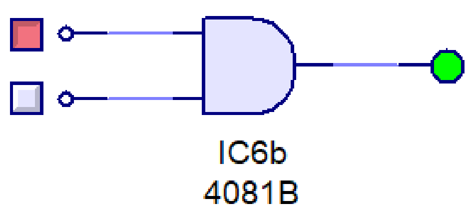
To combine the outputs of the XOR gate and the output of the astable, I decided to connect these two inputs to an AND gate which then goes into the clock input of the 4026. I then converted the AND gate into a NAND gate which I used when physically prototyping the scoreboard on a breadboard. But for the purpose of testing, I will be using the original layout with the single AND gate.* I included an AND gate so that the clock input of the 4026 is only triggered when both the astable and the monostable are HIGH and so that nothing happens when either one of them (and both) are LOW - otherwise the time period calculation I did earlier wouldn't work and hence be redundant. I expect the AND gate to provide a logic level of 1 only when both the astable and the output of the XOR gate is HIGH.

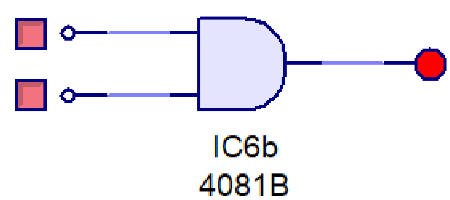
*The NAND gate conversion can be seen in the diagram below.





Subsystem circuit diagram





Testing

Test conditions	Input from XOR	Input from astable	Expected output	Actual output	Pass/Fail
Low XOR output and astable is in time low phase (passed falling edge)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
Low XOR output and astable is in time high phase (passed rising edge)	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
High XOR output and astable is in time low phase (passed falling edge)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
High XOR output and astable is in time high phase (passed rising edge)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Pass

Report on performance

As shown by the table above, the results are consistent with my expected outcomes stated in my specification where the output is only HIGH when both the output from the XOR gate and the astable is high (i.e. in the time high phase) and the output is LOW for all other scenarios. However, since this is even less likely than triggering 2 monostables by accident, this outcome isn't problematic to how the scoreboard works. Also, we can see that the table above matches with the truth table for a typical AND gate.

A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

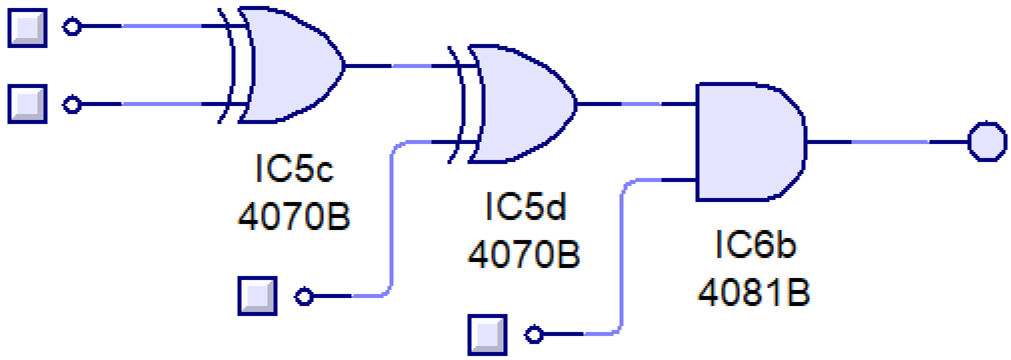
SUBSYSTEMS 5, 6 - 3 INPUT XOR GATE + AND GATE (DIGITAL)

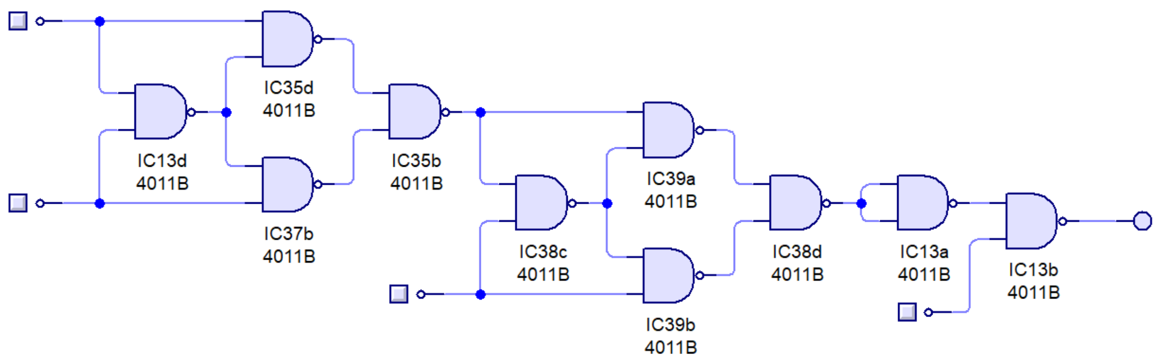
Specification

To make sure the entire logic of the scoreboard is working, I decided to combine both subsystems 5 and 6 together and tested this as a whole. The same reasons as to why I used this logic applies from the specification that I included for each separate subsystem (please see previous 2 subsystems for more information) but my expectations are slightly different as both logic components are now combined. Therefore, I expect the output of this logic to be HIGH in the 2 following scenarios:

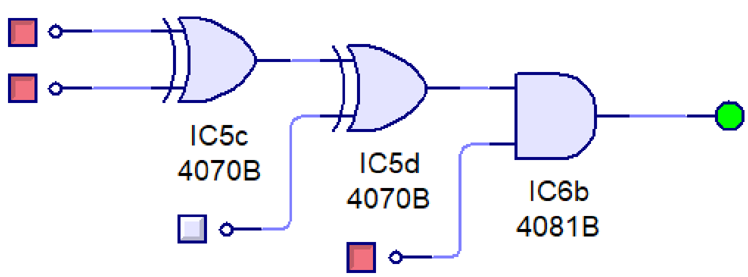
1. When only one monostable is triggered and the astable is in its time high phase.
2. When all three monostables are triggered at once and the astable is in its time high phase

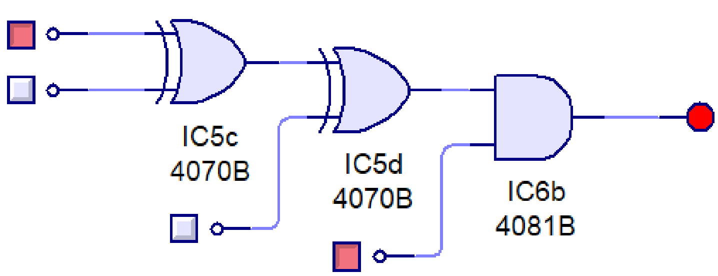
Since the output of this logic, as a whole, connects to the clock input of a 4026, a HIGH output includes the relevant number of rising edges needed for the value on the 7-segment to increment by a certain amount. The testing for this can be seen in the next few pages. A diagram of the original layout and the NAND layout can be seen below.





Subsystem circuit diagram





Testing

Test conditions	Input 1 run	Input 4 runs	Input 6 runs	Input from astable	Expected output	Actual output	Pass/Fail
No buttons pressed and astable low	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
No buttons pressed and astable high	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
6 runs button pressed and astable low	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
6 runs button pressed and astable high	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Pass
4 runs button pressed and astable low	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
4 runs button pressed and astable high	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Pass
4 and 6 runs buttons pressed and astable low	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
4 and 6 runs buttons pressed and astable high	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass

1 run button pressed and astable low	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
1 run button pressed and astable high	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Pass
1 and 6 runs buttons pressed and astable low	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
1 and 6 runs buttons pressed and astable high	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
1 and 4 runs buttons pressed and astable low	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
1 and 4 runs buttons pressed and astable high	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
1, 4 and 6 runs buttons pressed and astable low	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 0 (0V)	Logic level 0 (0V)	Logic level 0 (0V)	Pass
1, 4 and 6 runs buttons pressed and astable high	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Logic level 1 (9V)	Pass

Report on performance

The results show that the 4026 only receives a minimum of one rising edge at its clock input (and up to 6 based on the button pressed) in the following scenarios:

1. When only one monostable is triggered and the astable is in its time high phase.
2. When all three monostables are triggered at once and the astable is in its time high phase.

Again, as all three monostables being triggered is even less likely than triggering 2 monostables by accident, this scenario outcome isn't problematic to how the scoreboard works.

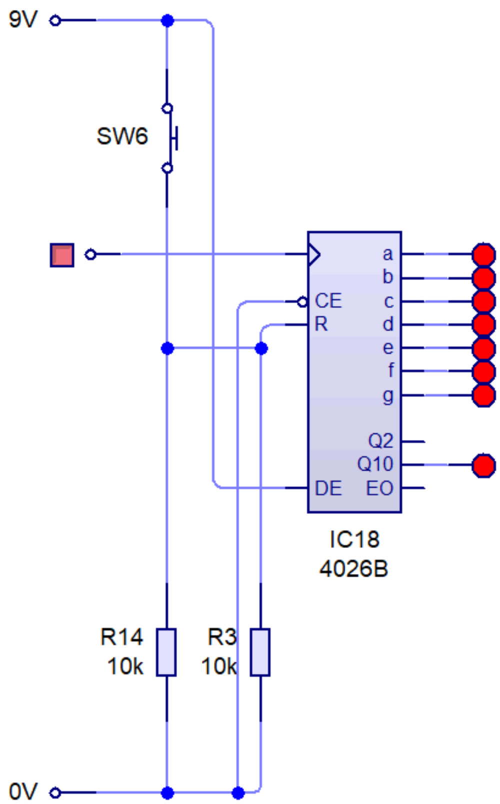
SUBSYSTEMS 7, 9, 11, 17, 20 - ONES (RUNS AND WICKETS), TENS AND HUNDREDS COLUMN 4026s + RESET SWITCH (DIGITAL)

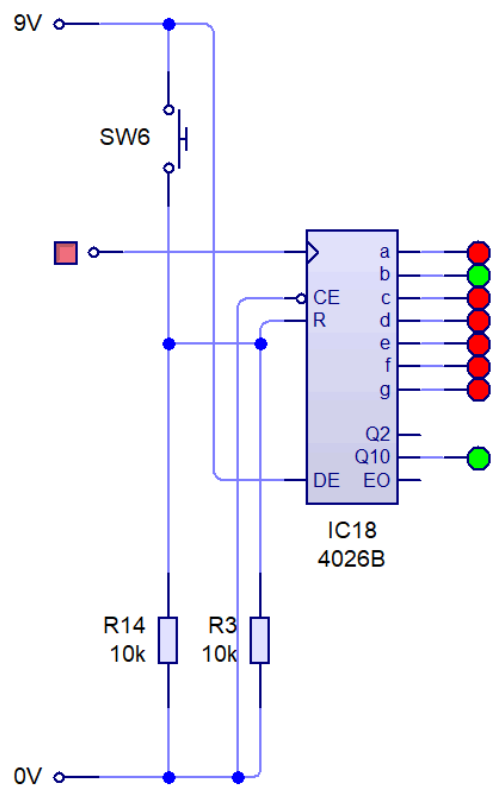
Specification

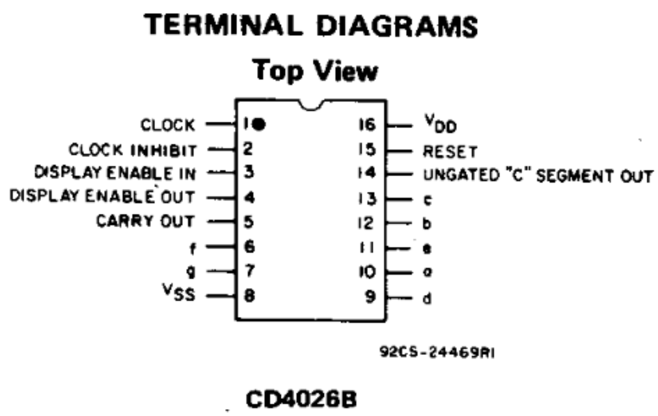
As per the specification, the scoreboard should consist of five 7-segment displays and to drive these, I decided to use a 4026 which consists of both a decade counter and a decoder and driver. Since one of my five 7-segment displays is a dash, only 4 of the 7-segment displays will require a 4026 and subsystems 7, 9, 11 and 17 make up the four 4026s I used for the scoreboard. The current through each segment of the 7-segment display should also be 4mA ± 0.5mA which means that the current at the output pins of the 4026s which then connect to their respective 7-segment display must be 4mA ± 0.5mA. There should also be 2 scores visible where one is for runs and the other is for wickets. Of the five 7-segment displays I will be using, 3 of them are for the runs score, 1 is for the dash and 1 is for the wickets score. As a result, three 4026s have been allocated for the runs score and one for the wickets score. The connections for the 4026s are identical for both the runs and wickets score so the below testing applies to the 4026s responsible for both scores. Lastly, the runs scoreboard should be able to allocate 3-digit scores which has already been satisfied as I am using three 7-segment displays for the runs score and hence three 4026s for the runs score. Also, I want the divide-by-10 output pin to be HIGH when the score on the ones and tens digit passes from 9 to 0 (when segments a, b, c, d, e, f are high). As for the reset switch, the reset button should reset both scores so that they both read 0 and should make up one of the 5 buttons mentioned in the specification.

Subsystem circuit diagram

I have connected the pins of the 4026s as shown by the datasheet where the clock enable and reset pins are active low and the display enable pin is active high. As a result, I connected the display enable pin to power and the clock enable pin to ground along with the reset pin via a 10kΩ resistor as I would need it for the reset switch to work.







Testing

Test conditions	Expected output	Actual output value	Pass/Fail	Corresponding value shown on 7-segment
Initial start up	All pins except g have a current of 4mA ± 0.5mA	All pins except g have a current of 3.83mA	Pass	0
First rising edge	Pins b and c have a current of 4mA ± 0.5mA	Pins b and c have a current of 3.83mA	Pass	1
Second rising edge	All pins except c and f have a current of 4mA ± 0.5mA	All pins except c and f have a current of 3.83mA	Pass	2
Third rising edge	All pins except e and f have a current of 4mA ± 0.5mA	All pins except e and f have a current of 3.83mA	Pass	3
Fourth rising edge	All pins except a, d and e have a current of 4mA ± 0.5mA	All pins except a, d and e have a current of 3.83mA	Pass	4
Fifth rising edge	All pins except b and e have a current of 4mA ± 0.5mA	All pins except b and e have a current of 3.83mA	Pass	5
Sixth rising edge	All pins except b have a current of 4mA ± 0.5mA	All pins except b have a current of 3.83mA	Pass	6
Seventh rising edge	Pins a, b and c have a current of 4mA ± 0.5mA	Pins a, b and c have a current of 3.83mA	Pass	7
Eighth rising edge	All pins have a current of 4mA ± 0.5mA	All pins have a current of 3.83mA	Pass	8
Ninth rising edge	All pins except e have a current of 4mA ± 0.5mA	All pins except e have a current of 3.83mA	Pass	9
Tenth rising edge	Same as initial start up but Q10 high	Same as initial start up and Q10 high	Pass	0
Reset switch, SW6, pressed	All pins except g have a current of 4mA ± 0.5mA	All pins except g have a current of 3.83mA	Pass	0

Report on performance

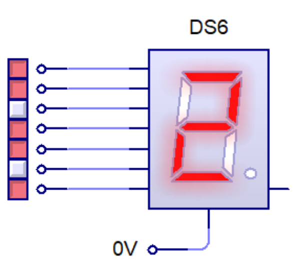
The results show that the 4026 correctly cycles through 0 to 9 as required with the current through each segment being 3.83mA. Also, when the reset switch is pressed, all the segments except g were on which indicates 0 on a 7-segment display so the subsystem(s) works as expected. And to note, the test involving the Q10 output is only for the 4026 responsible for the ones and tens digits for the runs score as they have been connected to the clock input of the 4062 for the 10s and 100s digits, respectively which causes the relevant digits to increase when the previous digit crosses from 9 to 0. Also, as the 4026 works by making the relevant segments high, a common cathode 7-segment display must be used for the scoreboard to work.

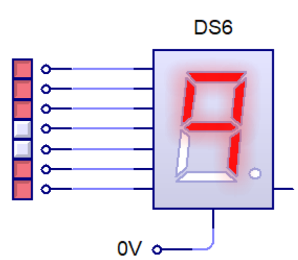
SUBSYSTEMS 8, 10, 12, 18 - ONES (RUNS AND WICKETS), TENS AND HUNDREDS COLUMN 7-SEGMENT DISPLAYS

Specification

As per the specification, the scoreboard should consist of five 7-segment displays and the 7-segment displays being tested here make up four of them as the last 7-segment display is for the dash. The current through each segment of the 7-segment display should also be 4mA ± 0.5mA which means that the current at the input pins of the 7-segment display must be 4mA ± 0.5mA. There should also be 2 scores visible where one is for runs and the other is for wickets. Of the five 7-segment displays I will be using, 3 of them are for the runs score, 1 is for the dash and 1 is for the wickets score. Lastly, the runs scoreboard should be able to allocate 3-digit scores which has already been satisfied as I am using three 7-segment displays for the runs score. To add, the logic gate inputs in the diagram below represent the output from the respective 7-segment display's 4026.

Subsystem circuit diagram





Testing

Test conditions	Expected output	Actual output	Pass/Fail
All 4026 output pins high except g	0 displayed with current in all pins except g being 4mA ± 0.5mA	0 displayed with current in all pins except g being 3.83mA	Pass
Only 4026 output pins b and c high	1 displayed with pins b and c having a current of 4mA ± 0.5mA	1 displayed with pins b and c having a current of 3.83mA	Pass
All 4026 output pins high except c and f	2 displayed with current in all pins except c and f being 4mA ± 0.5mA	2 displayed with current in all pins except c and f being 3.83mA	Pass
All 4026 output pins high except e and f	3 displayed with current in all pins except e and f being 4mA ± 0.5mA	3 displayed with current in all pins except e and f being 3.83mA	Pass
All 4026 output pins high except a, d and e	4 displayed with current in all pins except a, d and e being 4mA ± 0.5mA	4 displayed with current in all pins except a, d and e being 3.83mA	Pass
All 4026 output pins high except b and e	5 displayed with current in all pins except b and e being 4mA ± 0.5mA	5 displayed with current in all pins except b and e being 3.83mA	Pass
All 4026 output pins high except b	6 displayed with current in all pins except b being 4mA ± 0.5mA	6 displayed with current in all pins except b being 3.83mA	Pass
Only 4026 output pins a, b and c high	7 displayed with pins a, b and c having a current of 4mA ± 0.5mA	7 displayed with pins a, b and c having a current of 3.83mA	Pass
All 4026 output pins high	8 displayed with current through all pins being 4mA ± 0.5mA	8 displayed with current through all pins being 3.83mA	Pass
All 4026 output pins high except e	9 displayed with current in all pins except e being 4mA ± 0.5mA	9 displayed with current in all pins except e being 3.83mA	Pass
All 4026 output pins high except g	0 displayed with current in all pins except g being 4mA ± 0.5mA	0 displayed with current in all pins except g being 3.83mA	Pass

Report on performance

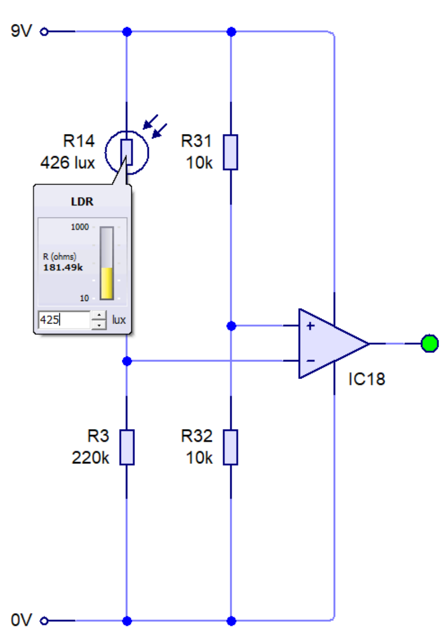
The results show that the 7-segment display(s) correctly cycles through 0 to 9 as required with the current through each segment being 3.83mA. To note, I connected the common pin of the 7-segment display to ground as the 7-segment display used for testing is a common cathode. So subsystem(s) works as expected.

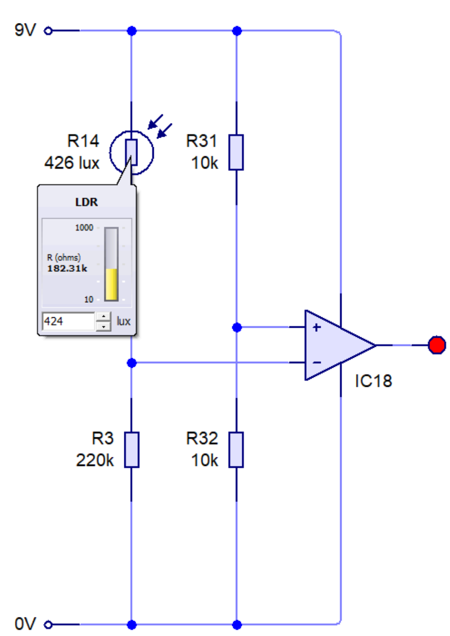
SUBSYSTEM 13 - LIGHT SENSOR (ANALOGUE)

Specification

I decided to design the light sensing subsystem with an LDR connected to an op-amp such that the output of the op-amp is high when the light level falls below 425 lux ± 10 lux, which I found was to be the approximate light level at sunset. As the output of the op-amp connects to a row of LEDs, the idea behind this subsystem is that the LEDs turn on at sunset as the light level falls from above 425 lux to below 425 lux.

Subsystem circuit diagram





Testing

Test conditions	Light level / lux	Expected output	Actual output value	Pass/Fail
Input voltage = 4.74V	450	Logic level 0 (0V)	Logic level 0 (0V)	Pass
Input voltage = 4.58V	435	Logic level 0 (0V)	Logic level 0 (0V)	Pass
Input voltage = 4.49V	426	Logic level 0 (0V)	Logic level 0 (0V)	Pass
Input voltage = 4.48V	425	Logic level 0 (0V)	Logic level 0 (0V)	Pass
Input voltage = 4.47V	424	Logic level 1 (9V)	Logic level 1 (9V)	Pass
Input voltage = 4.38V	415	Logic level 1 (9V)	Logic level 1 (9V)	Pass
Input voltage = 4.23V	400	Logic level 1 (9V)	Logic level 1 (9V)	Pass

Report on performance

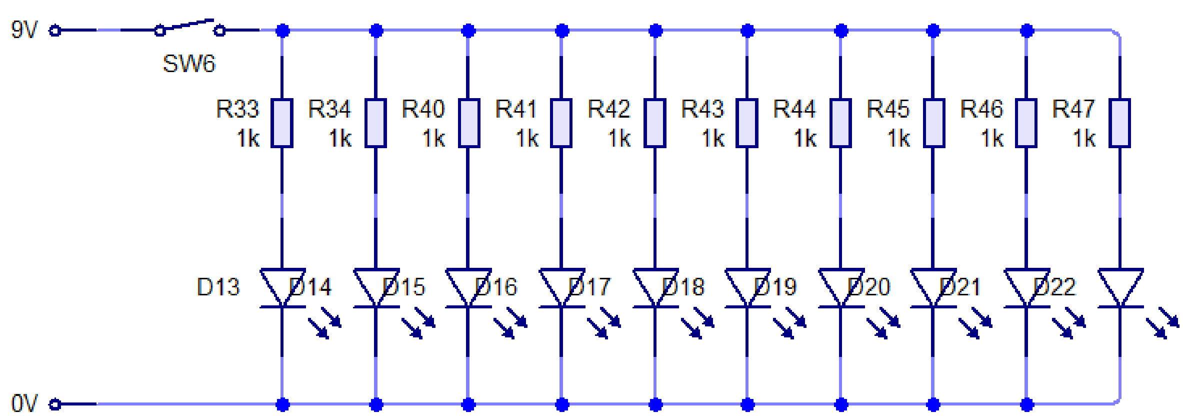
I designed the light sensing subsystem by making two potential dividers where one included the LDR and the other included two 10k resistors so that V_{ref} for the op-amp was theoretically 4.5V but was actually 4.48V when tested. The inverting input connects to the LDR potential divider and the non-inverting input connects to the 10k potential divider. The arrangement of the LDR potential divider is such that the value of R3 in the diagram below is just enough for the op-amp to stay low until the light intensity decreases below 425 lux, which causes the resistivity of the LDR to increase, decreasing the voltage across R3. As this voltage passes 4.48V, the output of the op-amp switches from a logic level of 0 to a logic level of 1, which is then connected to the row of white LEDs. As shown by the results, I tested the subsystem at the following light levels: 435 lux, 425 lux, 415 lux (and more), which include the upper and lower bounds in my uncertainty of 425 lux ± 10 lux. I found that the output of the op-amp is high only at 415 lux (and at 424 lux which I also tested) which is what I expected, so the subsystem works as expected.

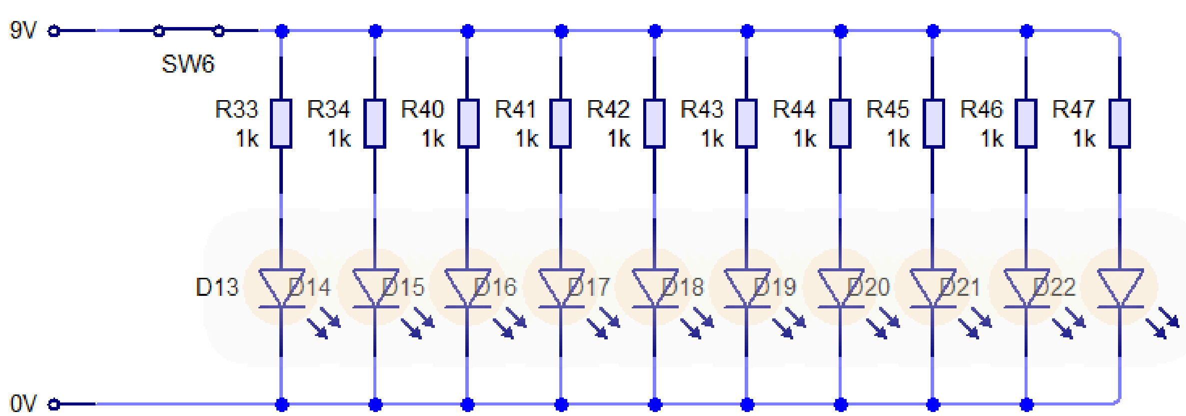
SUBSYSTEM 18 - WHITE LED ROW

Specification

As per the specification, there should be a row of white LEDs which turn on when it gets dark and the power dissipated by each LED should be 5mW ± 0.5mW. I also added a 1kΩ resistor to protect the LED from high currents. To test this subsystem, I decided to simulate a high output voltage from the output of the op-amp, I am using a closed switch connected to a 9V power supply and to simulate a low output voltage from the output of the op-amp, I am using an open switch connected to 9V. The cathode of the LEDs connect to ground.

Subsystem circuit diagram





Testing

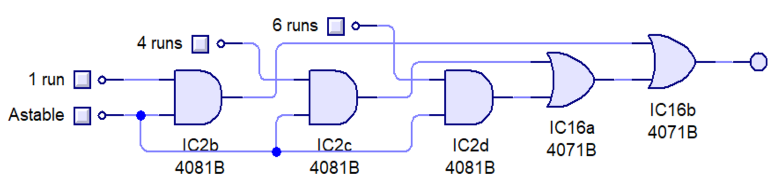
Test conditions	Expected output	Actual output value at collector	Pass/Fail
Switch off, output of op-amp is low as light level ≥ 425 lux	LEDs off, power dissipated by each LED of 0W	LEDs off, power dissipated by each LED of 0W	Pass
Switch on, output of op-amp is high as light level < 425 lux	LEDs on, power dissipated by each LED of 5mW ± 0.5mW	LEDs on, power dissipated by each LED of 5.17mW	Pass

Report on performance

The results show that when the switch is closed, the input voltage is at logic level 1 (9V) so the row of LEDs is on and when the switch is open, the input voltage is at logic level 0 (OV) so the row of LEDs is off.

ALTERNATIVE SUBSYSTEM DESIGNS

1. Subsystems 5 and 6 could be replaced with 3 AND gates and a 3-input OR gate in the following arrangement:



By looking at the truth table for this arrangement below, we can see that the output is high in the same cases as the original logic diagram I used, but also has high outputs when 2 or more buttons and the astable are high. This would mean that if 2 buttons were pressed by accident, the score would increment - which is slightly problematic as this is far more likely to occur compared to the one odd scenario in my original logic where the output was high also when all three buttons were pressed. But, unlike the original design, this logic can be simplified using NAND gates, which can be seen below the truth table, and reduces both the number of chips and gates used. However, to ensure that the score doesn't increment when even 2 buttons are pressed, I decided to use my original design.

1 run	4 runs	6 runs	Astable	Output
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

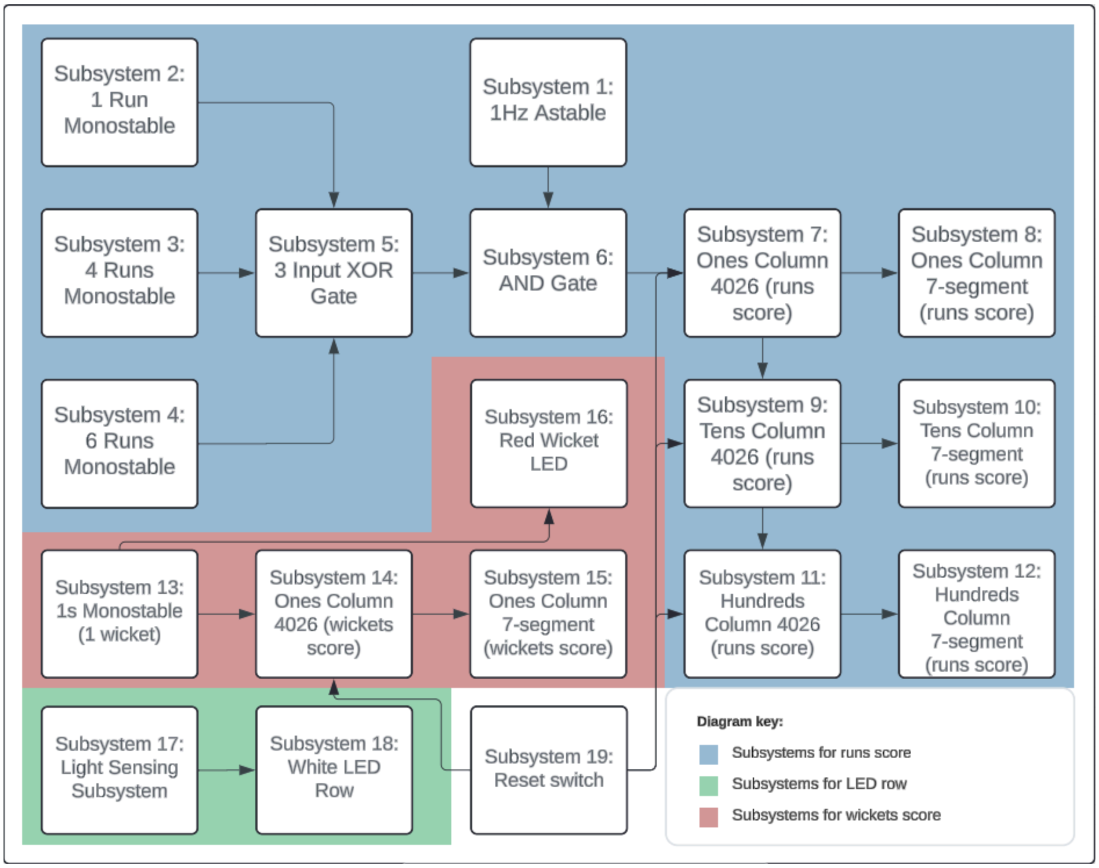
2. Subsystem 1 could be replaced with a 1Hz crystal oscillator which would make the frequency output more accurate and hence closer to 1Hz. The duty cycle of the crystal oscillator would also be exactly, if not very close to, 50% which would make calculating the time periods of the monostables much easier and since they use less power, they would be more economical. However, although crystal oscillators aren't very expensive, the ICs required to divide the 32kHz output into a 1Hz input is much more expensive so this wouldn't be beneficial for a one-off scoreboard and their simple design means that they can be tweaked easily. So, if the current system was to be improved and a higher frequency output from the astable was required, this wouldn't be possible to do if a crystal oscillator is used as they are of fixed frequencies. Therefore, based on these factors, I felt that it

would be better to continue with an astable instead of a fixed-frequency crystal oscillator.

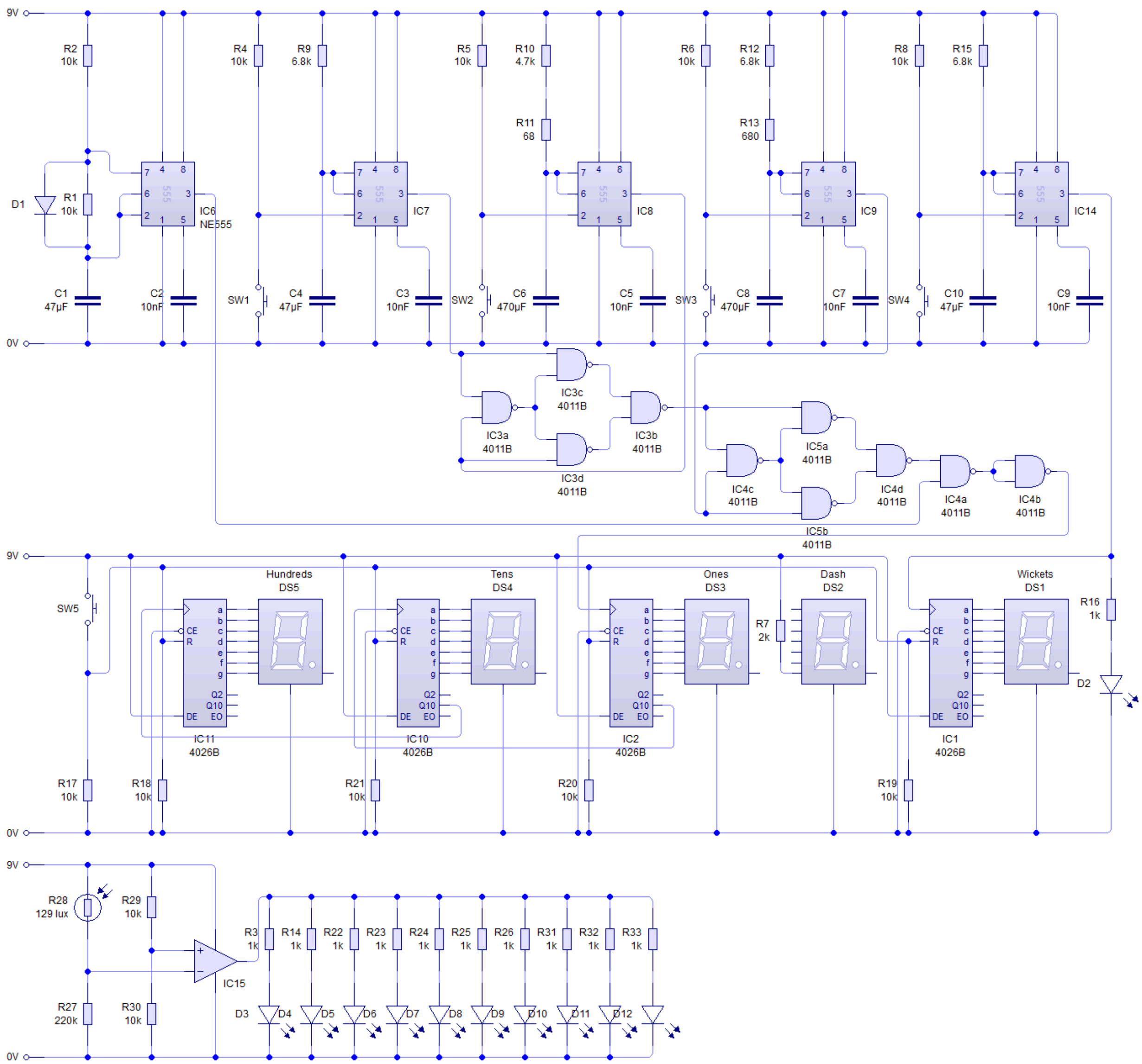
3. Also, a new subsystem could be introduced - an NPN transistor. This would be used to amplify the current through the row of white LEDs, making them brighter. If the specification was modified so that the current through the LEDs must be of a higher value, we can connect the output of the op-amp to the base of the transistor, the cathodes of the LEDs connected to the collector with their load resistors and the emitter connected to ground. We can then use the formula $I_C = h_{FE}*I_B$ to calculate the required base current from the op-amp for the transistor to function, altering I_B with a resistor after the output of the op-amp. However, as my user was satisfied with the brightness of the LEDs without amplification, I decided to omit the NPN transistor.
4. The 7-segment displays could also be replaced with a singular LCD display which would make the system much more compact. Also, since there is more space on an LCD display, more characters can be displayed such as how many balls are left to be bowled in an over, how many runs each batsman has scored etc. Lastly, LCD displays require slightly less power to function, especially if the information being displayed is static which would be the case for cricket matches as the games tend to last a long time and the score might be constant in between overs or if a test match is being played where the score typically changes much less compared to other formats. However, since 7-segment displays are significantly cheaper, are much more clearly visible in overcast weather and are relatively simple to hook up to a decade counter and decoder and driver, I decided to use 7-segment displays.
5. Lastly, one could argue that it would be much easier to replace the 4026s and even the system as a whole with a microcontroller such as a PIC16F84A as this would mean that changes to the scoreboard’s design can be made much more quickly (e.g. how many balls left in an over) and the process itself would be much easier than tweaking the RC values on the monostables. However, again, microcontrollers are a much more expensive improvement and cost more compared to 4026s. Also, the chance of errors when programming the microcontrollers (e.g. syntax errors) are much more likely to occur compared to 4026s. Therefore, as integrating a microcontroller is a much more complex way of solving the existing problem and the above, I decided to use 4026s and logic with monostables and astables, instead.

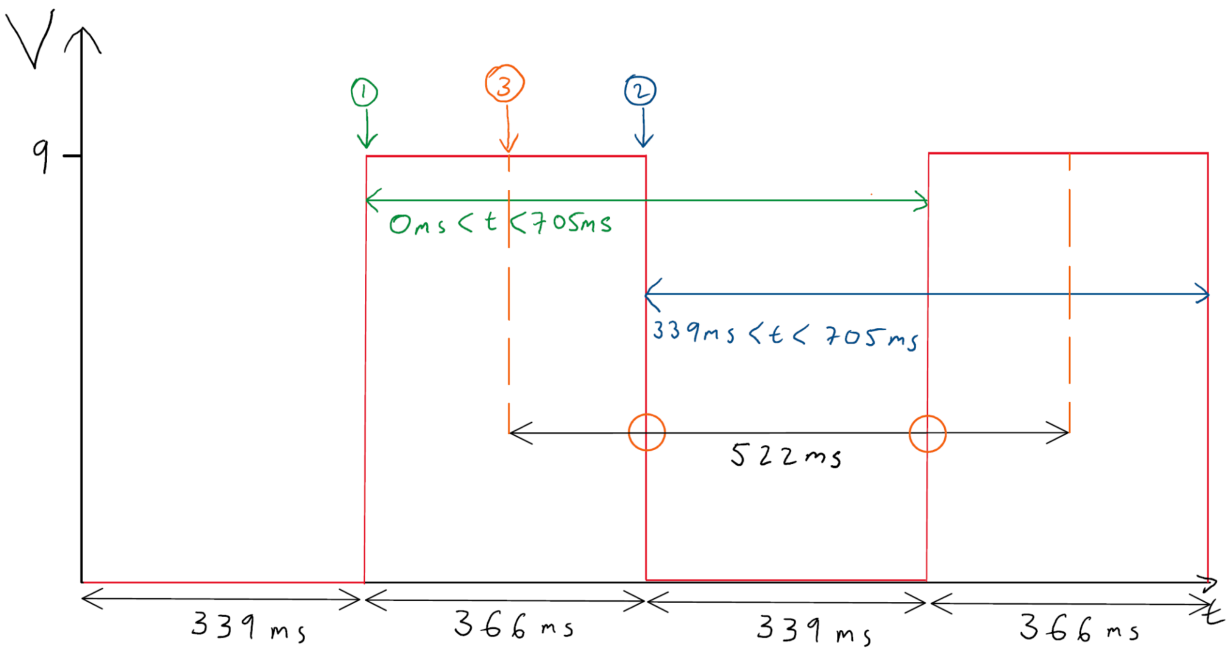
SYSTEM REALISATION

SYSTEM BLOCK DIAGRAM OF COMPLETED SYSTEM

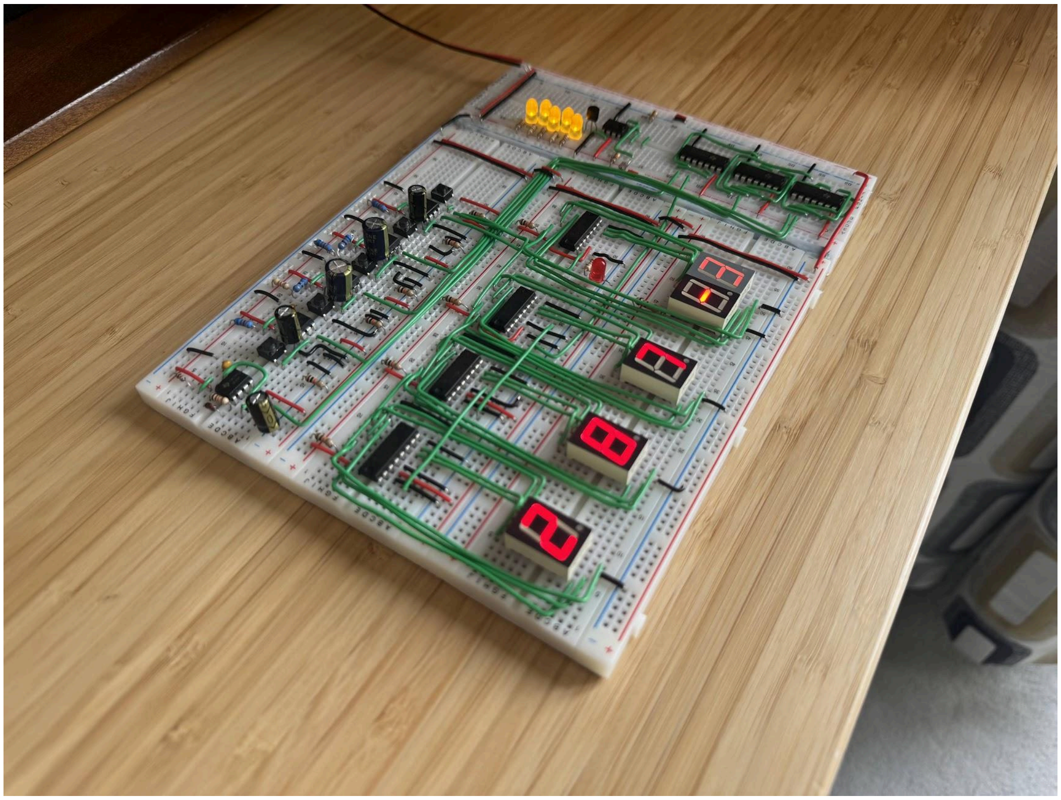


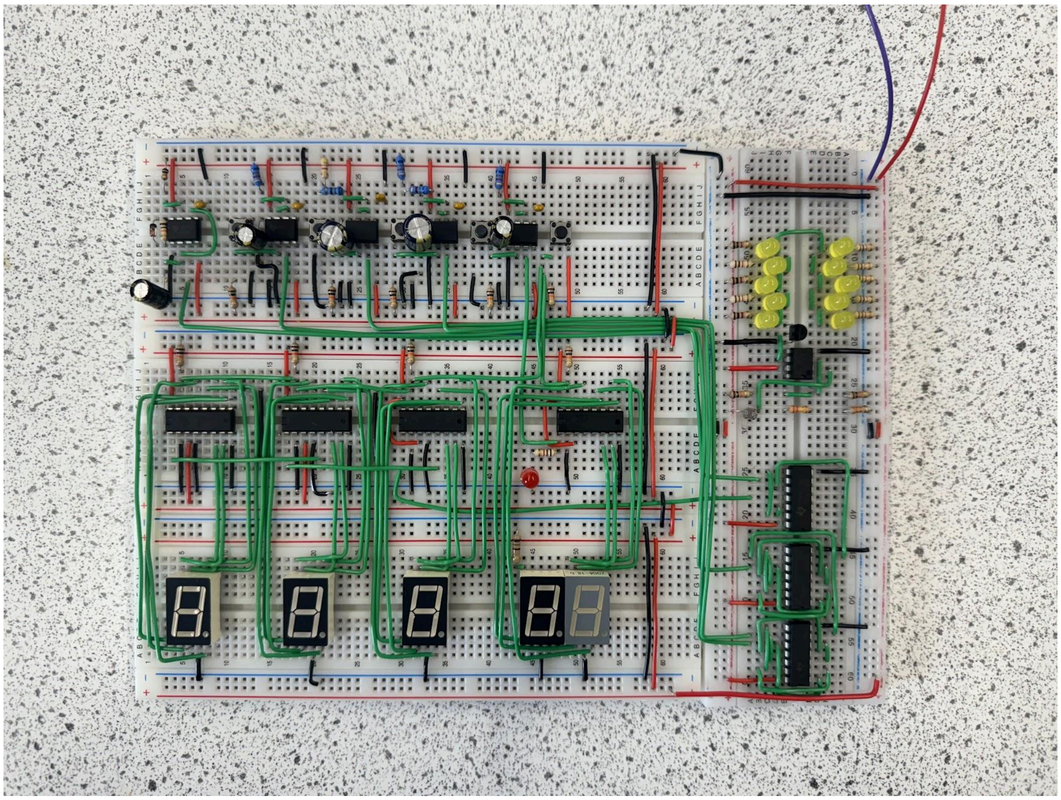
CIRCUIT DIAGRAM OF COMPLETED SYSTEM

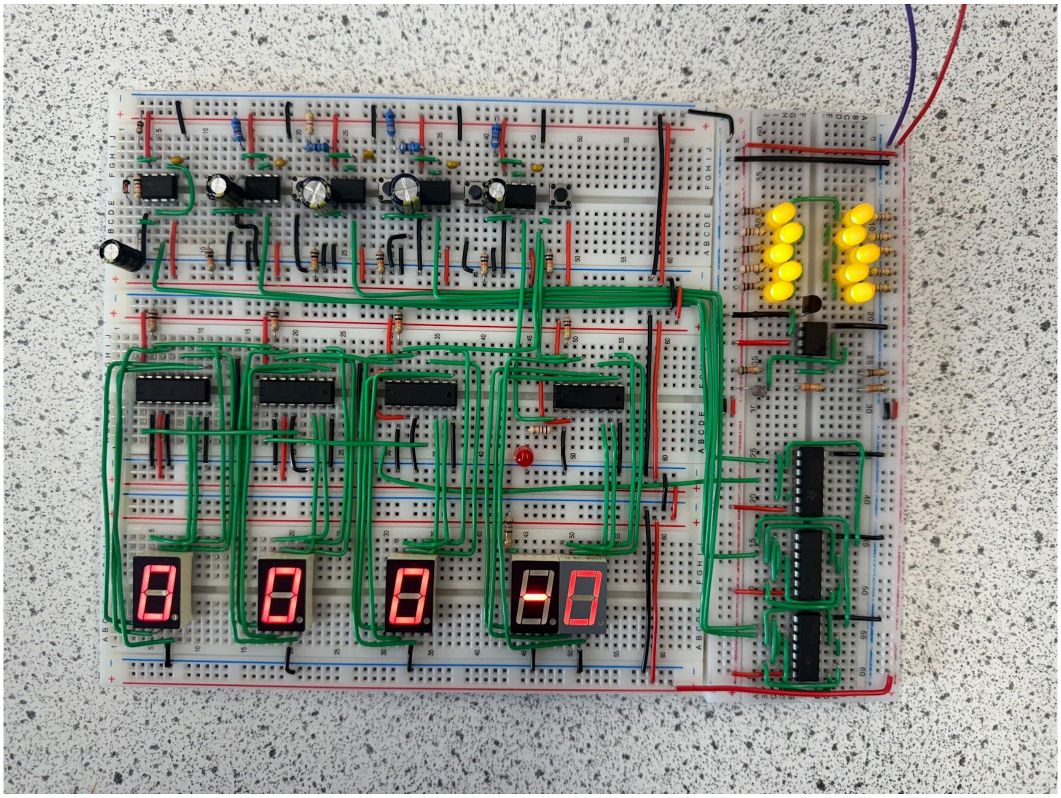


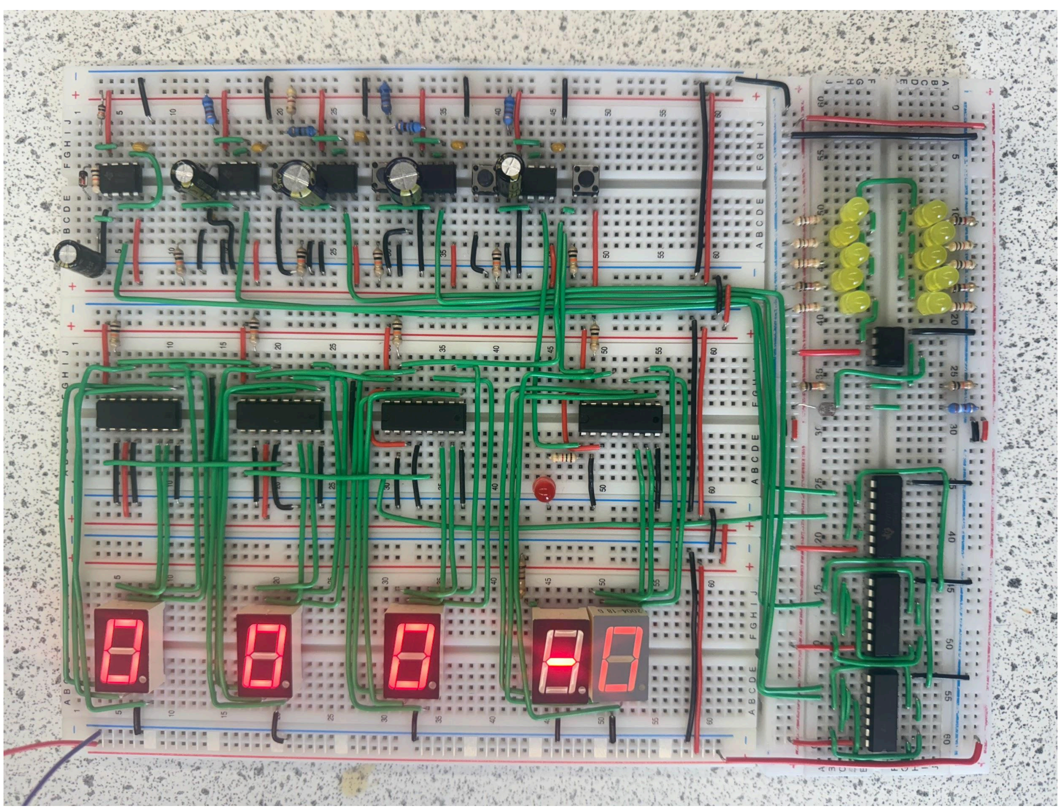

V
9
1
3
2
0ms < t < 705ms
339ms < t < 705ms
522ms
339 ms
366 ms
339 ms
366 ms
t

PHYSICAL CIRCUIT LAYOUT PROTOTYPED ON BREADBOARDS

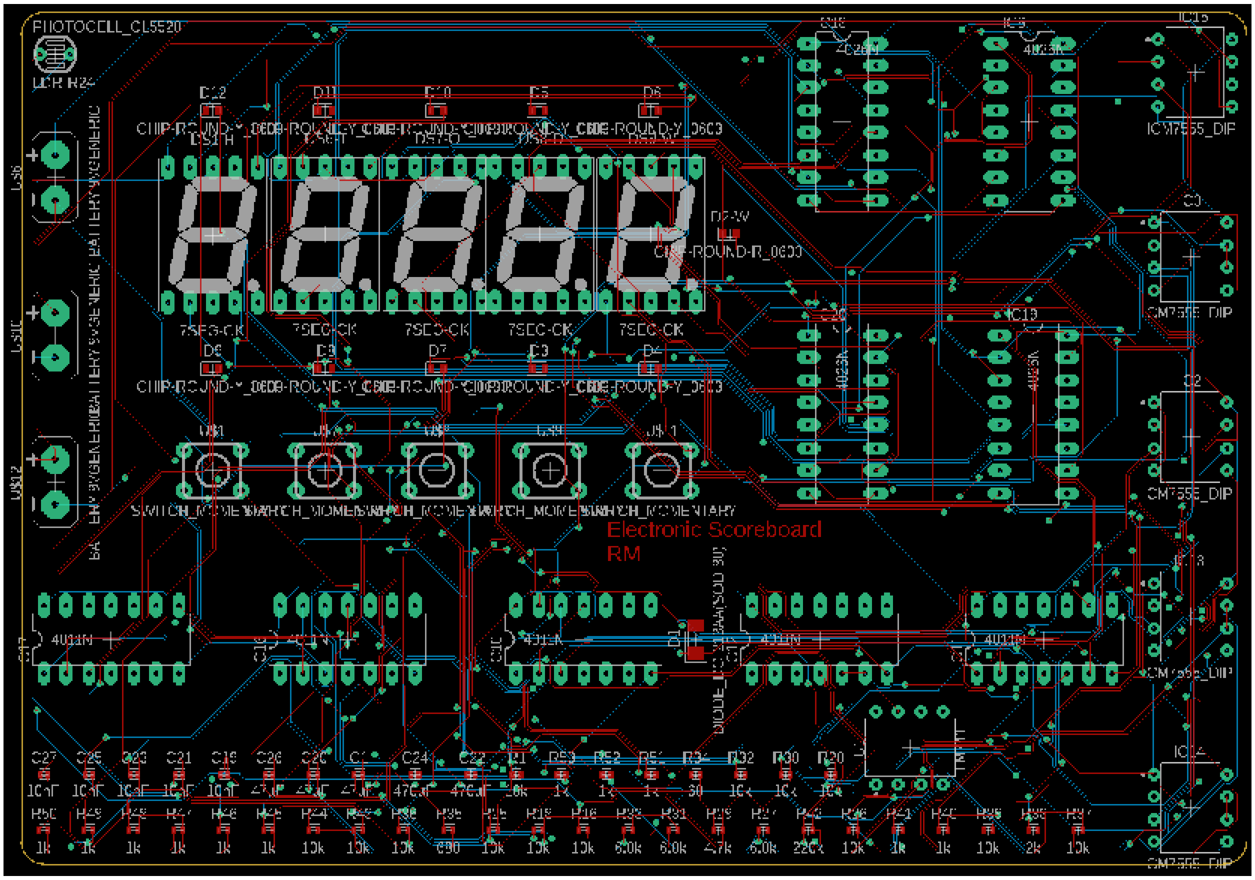


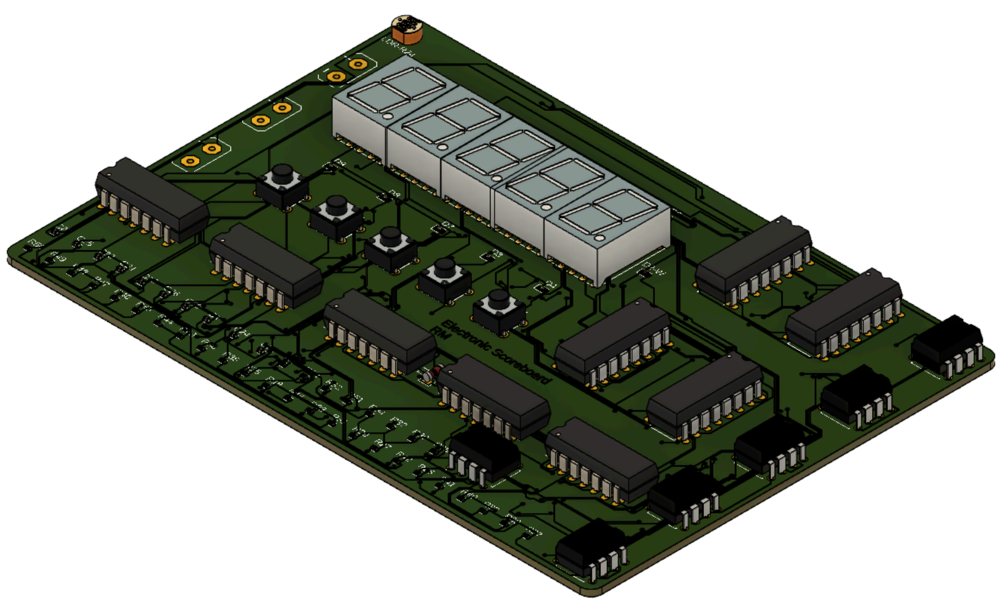






CIRCUIT CONVERSION AND PCB DESIGN ON FUSION 360





COMPONENT LIST

Component Name	Quantity	Component Name	Quantity
10K Resistor	13	Diode	1
10nF Capacitor	13	Op-amp	1
1K Resistor	1	LDR	1
200K Resistor	1	NE555	5
20K Resistor	1	Push-to-make switch	5
2K Resistor	1	Red LED	1
4.7K Resistor	1	White LED	10
4011B 2-Input NAND Gate	3		
4026B Decade Counter, Decoder and Driver	4		
47μF Capacitor	3		
470μF Capacitor	2		
6.8k Resistor	3		
68Ω Resistor	1		
680Ω Resistor	1		
7-Segment Display	5		

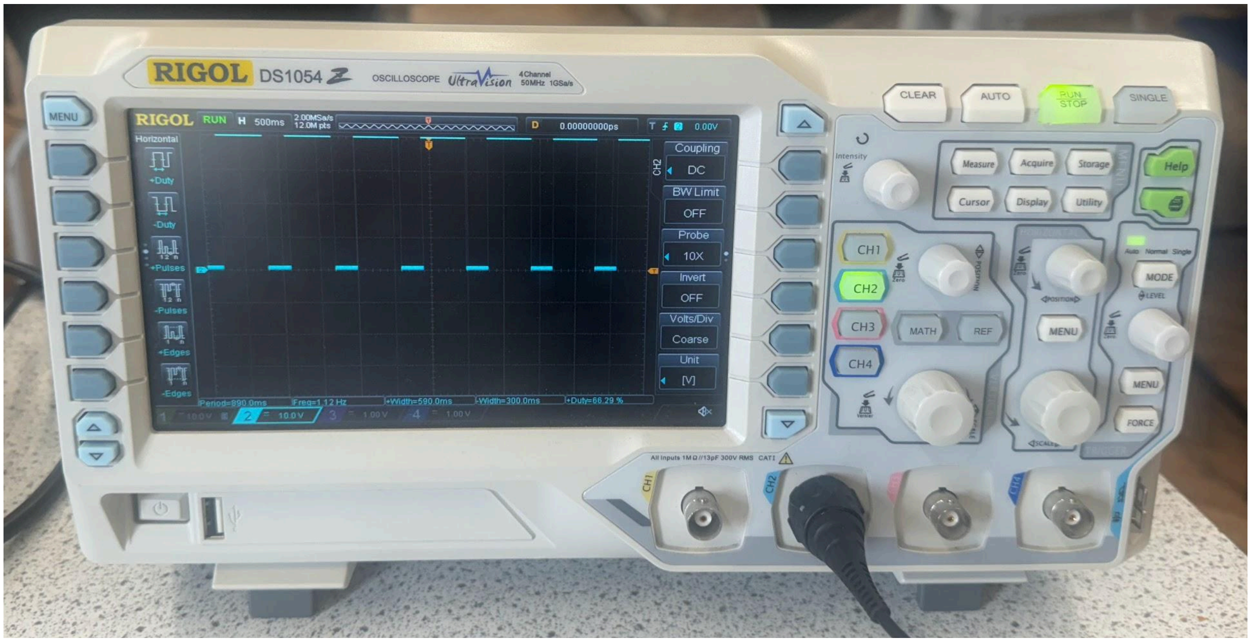
TESTING OF COMPLETE SYSTEM + PROOF OF TESTING

Test conditions	Expected output	Actual output	Pass/Fail
Start-up, no buttons pressed, light level ≥ 425 lux	Score shows 110-0, row of white LEDs off, power dissipated by each white LED is 0W	Score shows 110-0, row of white LEDs off, power dissipated by each white LED is 0W	Pass
Reset button pressed, light level ≥ 425 lux	Score shows 000-0, row of white LEDs off, power dissipated by each white LED is 0W	Score shows 000-0, row of white LEDs off, power dissipated by each white LED is 0W	Pass
1 run button pressed, light level ≥ 425 lux	Score shows 001-0, row of white LEDs off, power dissipated by each white LED is 0W	Score shows 001-0, row of white LEDs off, power dissipated by each white LED is 0W	Pass
4 runs button pressed, light level ≥ 425 lux	Score shows 005-0, row of white LEDs off, power dissipated by each white LED is 0W	Score shows 005-0, row of white LEDs off, power dissipated by each white LED is 0W	Pass
6 runs button pressed, light level ≥ 425 lux	Score shows 011-0, row of white LEDs off, power dissipated by each white LED is 0W	Score shows 011-0, row of white LEDs off, power dissipated by each white LED is 0W	Pass
1 wicket button pressed, light level ≥ 425 lux	Score shows 011-1, red LED on when button pressed, current through the red LED should be 7mA ± 0.5mA, row of white LEDs off, power dissipated by each white LED is 0W	Score shows 011-1, red LED on when button pressed, current through red LED is 6.78mA, row of white LEDs off, power dissipated by each white LED is 0W	Pass
Start-up, no buttons pressed, light level < 425 lux	Score shows 110-0, row of white LEDs on, power dissipated by each white LED should be 5mW ± 0.5mW	Score shows 110-0, row of white LEDs on, power dissipated by each white LED is 5.15mW	Pass
Reset button pressed, light level < 425 lux	Score shows 000-0, row of white LEDs on, power dissipated by each white LED should be 5mW ± 0.5mW	Score shows 000-0, row of white LEDs on, power dissipated by each white LED is 5.15mW	Pass

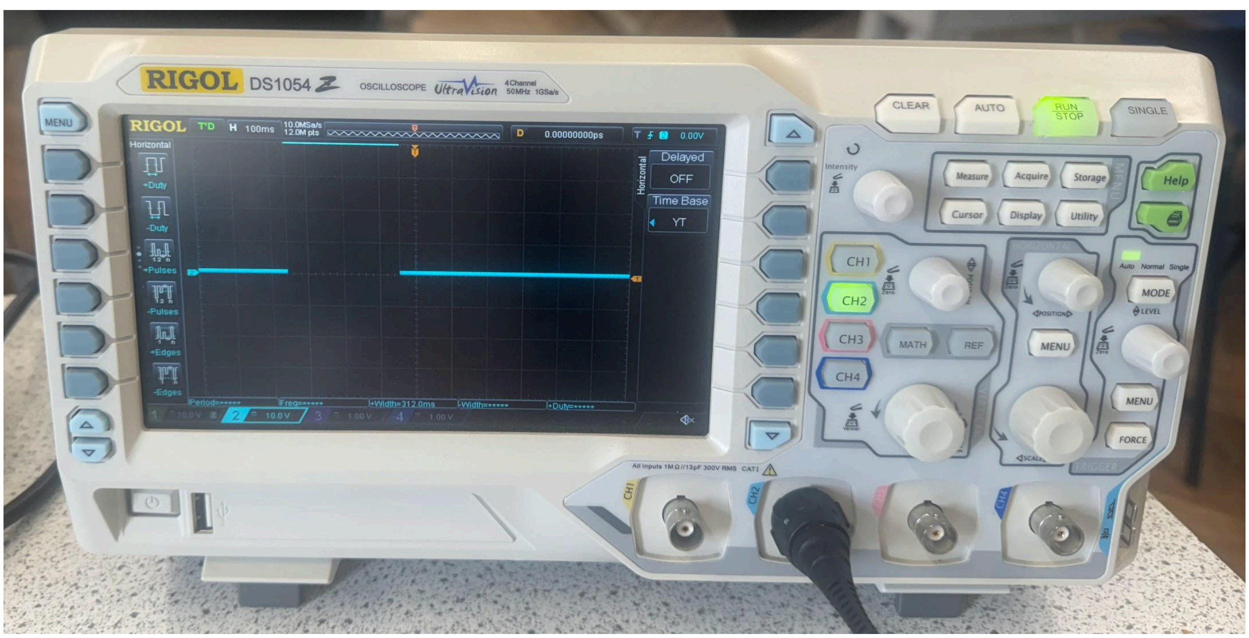
1 run button pressed, light level < 425 lux	Score shows 001-0, row of white LEDs on, power dissipated by each white LED should be 5mW ± 0.5mW	Score shows 001-0, row of white LEDs on, power dissipated by each white LED is 5.15mW	Pass
4 runs button pressed, light level < 425 lux	Score shows 005-0, row of white LEDs on, power dissipated by each white LED should be 5mW ± 0.5mW	Score shows 005-0, row of white LEDs on, power dissipated by each white LED is 5.15mW	Pass
6 runs button pressed, light level < 425 lux	Score shows 011-0, row of white LEDs on, power dissipated by each white LED should be 5mW ± 0.5mW	Score shows 011-0, row of white LEDs on, power dissipated by each white LED is 5.15mW	Pass
1 wicket button pressed, light level < 425 lux	Score shows 011-1, red LED on when button pressed, current through the red LED should be 7mA ± 0.5mA, row of white LEDs on, power dissipated by each white LED should be 5mW ± 0.5mW	Score shows 011-1, red LED on when button pressed, current through red LED is 6.78mA, row of white LEDs on, power dissipated by each white LED is 5.15mW	Pass

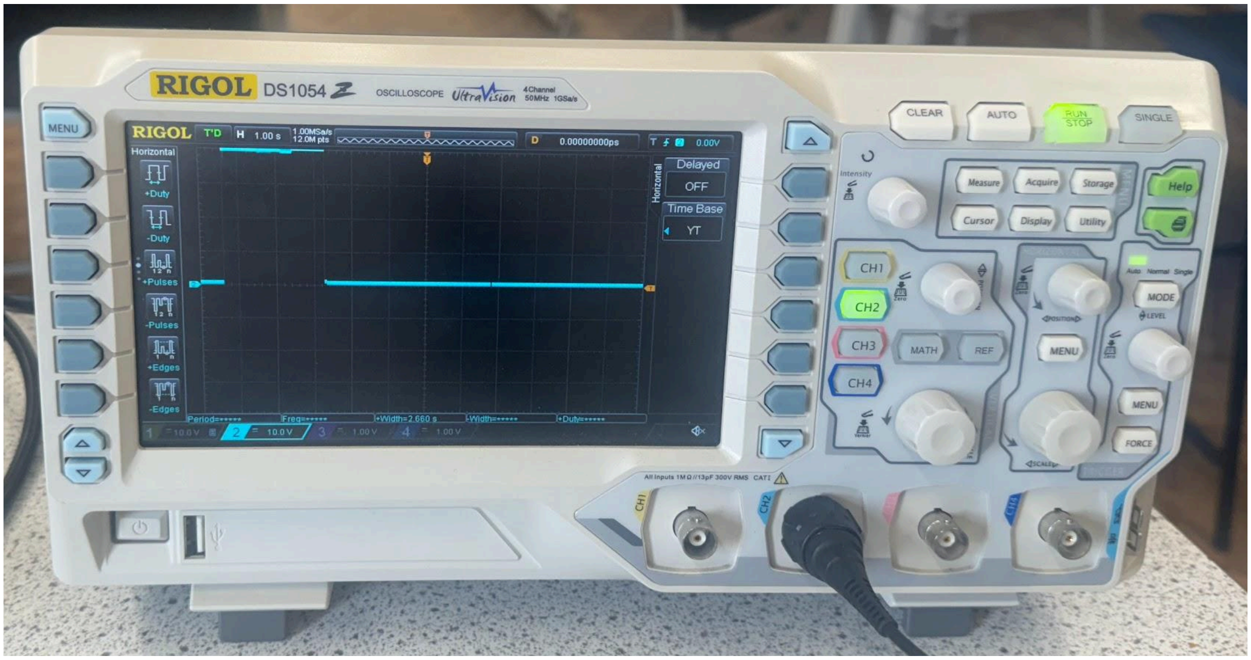
To measure wattage, I used a multimeter to find the voltage across the LEDs and then the current through the LEDs. I then multiplied these values to give wattage. In the same way, to measure the current through the red LED, I used a multimeter.

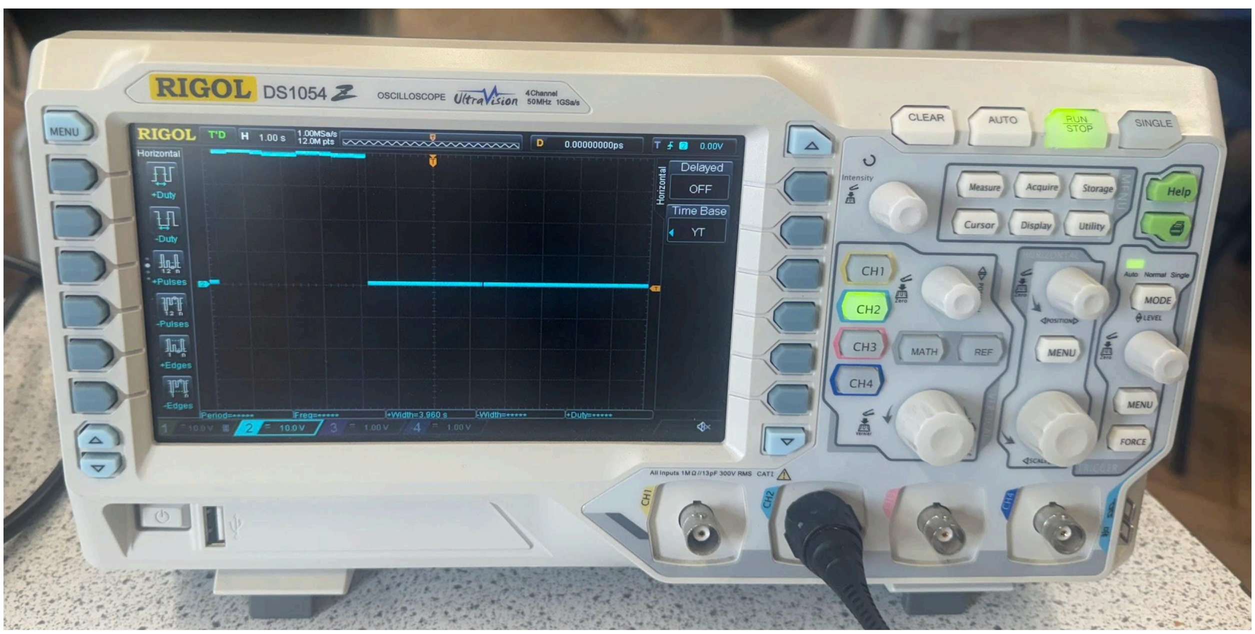
The oscilloscopes below show the frequency of the astable as 1.42Hz.

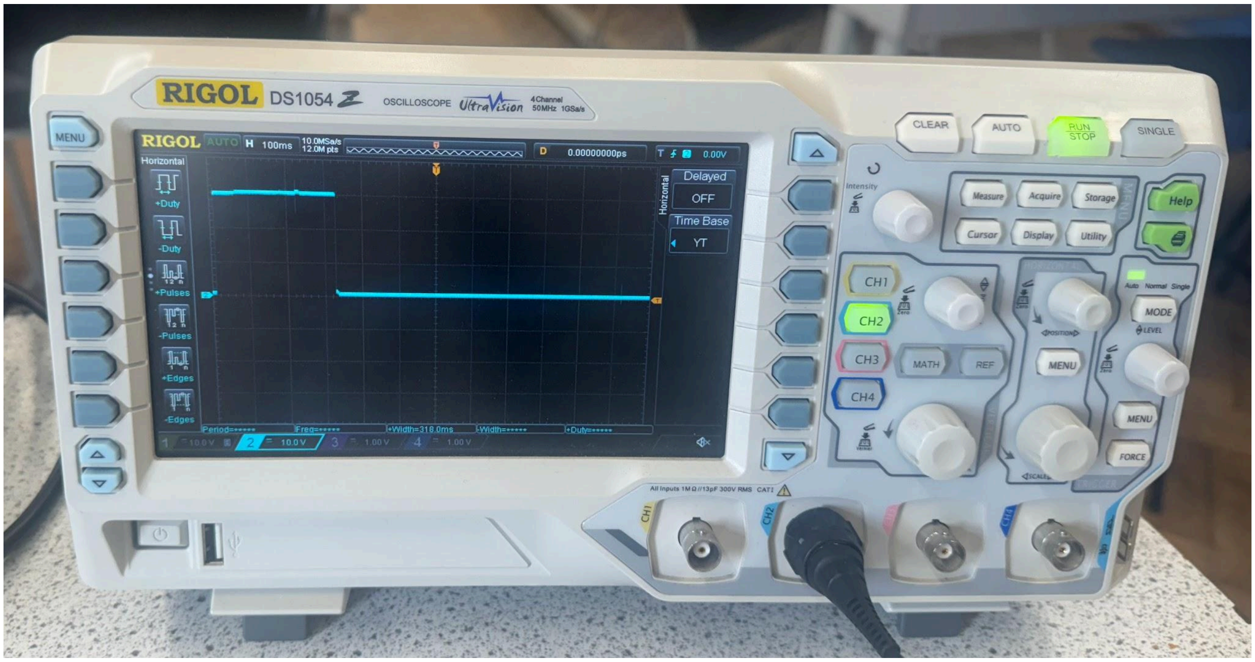


The oscilloscopes below show the time high of the 1 run monostable, 4 runs monostable, 6 runs monostable and 1 wicket monostable to be 312ms, 2.66s, 3.96s and 318ms respectively.




RIGOL
DS1054Z
OSCILLOSCOPE
UltraVision
CLEAR
AUTO
SINGLE
Measure
Acquire
Storage
Cursor
Display
Utility
Help
CH1
CH2
CH3
CH4
MATH
REF
MENU
MODE
FORCE


RIGOL
DS1054Z
OSCILLOSCOPE
UltraVision
CLEAR
AUTO
SINGLE
Measure
Acquire
Storage
Cursor
Display
Utility
Help
CH1
CH2
CH3
CH4
MATH
REF
MENU
MODE
FORCE



ANALYSIS OF RESULTS

As the results show that in every possible scenario, from start-up to end of scoring for the first team, the appropriate wattage is dissipated by the red LED and the appropriate current goes through the white LEDs, this shows that the system works as per the requirements given in the specification and is hence an acceptable design.

INTERFACING ISSUES

One of the interfacing issues I faced with my initial design of the 1 run 4026 was that the value on the 7-segment display would rapidly increase because the switch connected to the pull-up resistor would bounce. This is because once the switch was pressed, the metal contacts would vibrate and collide, causing the output to rapidly oscillate between high and low. And since 4026s are rising-edge-triggered, this meant that every time the switch was pressed, the score would increase very quickly and not one at a time. So to fix this, I decided to connect the output of the switch to a monostable and the output of the monostable goes into the clock input of the 4026. This way, when the switch is pressed, the monostable is triggered, causing the output of the 555 to be high for a certain time period which I calculated to be 351.56ms. And in this time period, since there is only one rising edge, the value on the 7-segment display only increases by 1. Another interfacing issue I faced was that my initial design of the light sensing subsystem would provide an analogue output to the row of LEDs rather than a binary/digital output which was what I required. This was because I initially only had a potential divider consisting of a LDR in series with a variable resistor and the output of the light sensing subsystem going into the row of LEDs was the potential difference across the variable resistor. This was not ideal as for many LDRs, the output voltage varies relative to their resistances. So to convert this analogue input into a digital output, I decided to use an op-amp and light sensing subsystem in a bridge arrangement so that the op-amp was constantly measuring the potential difference across both inputs and checking if this was below the threshold voltage. By doing this, the row of LEDs turn on when the voltage is below the threshold and are off when the voltage is above the threshold. Lastly, an interfacing issue I mentioned earlier is the monostable being triggered when the astable is partly through its time high phase. Initially, I calculated the time period of the monostable by using inequalities and considering 2 “extreme” scenarios (please see page 5 for more information). However, I realised that I missed out on a third “extreme” scenario where the monostable could be triggered when the astable was midway through its time high period. This caused the score to increase by 1 more than expected. So to fix this, for the one run astable in particular, I decided to make the time period of the monostable as close to the time low of the astable as I could so that it would last long enough to pass one rising edge but no more.

USER GUIDE

For the system to function, first, a power source must be provided to the circuit. This would ideally be a 9V battery. After that, the 7-segments would light up and based on testing, I saw that on start-up, the score would typically be 110-0. So to restart the score on 000-0, the reset switch must be pressed. From this point, the scoring for both wickets and runs can begin. If the team batting scores 1 run, the 1 run button can be pressed, if they score 4 runs, the 4 runs button can be pressed and if they score 6 runs, the 6 runs button can be pressed. For 2 and 3 runs, the best way to do this would be by pressing the 1 run button twice or thrice based on the runs scored. Since this scoreboard doesn’t account for the number of balls bowled in an over, it’s not possible to record wide or no balls or any extra runs given by the umpire. So instead, to ensure the correct runs score is displayed, the number of extra runs given can be treated as runs which can then be added by pressing one of the three buttons for the corresponding number of runs. In the same way, every time the team loses a wicket, the 1 wicket button can be pressed. When the number of wickets displayed on the scoreboard changes from 9 to 0, this means that the batting team is all out so the final score would then be recorded and the reset switch would be pressed again, causing the score to reset to 000-0 for the next team to start batting. The score should also stop when all the overs have been bowled, which would be signalled by the umpire on the field. The exact same process for scoring runs and losing wickets that applied to the first team batting also applies to the second team batting until the number of runs scored by the second team passes that of the first team or if the second team loses 10 wickets. After the match is finished, the scoreboard can be disconnected from the power supply until the next match.

EVALUATION

Performance Evaluation

When initially connected to a power supply, the score would read 110-0 as the 4026s that are responsible for the 10s and 100s digit for the runs score have their clock inputs connected to the Q10 (÷10) output of the 4026 connected to the 1s digit and 10s digit, respectively. And as the Q10 output is high for 0-4 and low for 5-9, this causes the 7-segment displays responsible for the 10s and 100s digit of the runs score to output a pulse, causing the value on the 7-segments to increase from 0 to 1. To solve this problem, I decided to use a reset switch which is connected in series with a pull-up resistor so that the voltage across the switch when pressed is 0V. This is necessary for the trigger pin of the 4026 as it is active low which means a logic level of 0 (0V) is needed for the 7-segment displays to reset back to 0. Once the score has been reset, the user can start scoring the number of runs and wickets scored by the team batting. This is done with an astable of frequency 1.42Hz and 3 monostables, one for 1 run, another for 4 runs and the last one for 6 runs. Since the time period of 1 cycle of the astable is 0.704s or 704ms, I had to calculate suitable time periods of the monostables such that the 1 run monostable's time period lasted 1 rising edge, the 4 run monostable lasted 4 rising edges and the 6 run monostable lasted 6 rising edges. As the output of these monostables connect to the clock input of 4026s, which increase the number displayed on the 7-segment displays by 1 whenever a rising edge is detected, this causes the runs score to increase by 1, 4 or 6 based on which button is pressed. And since the circuitry of the 1 run monostable can be used to increment the number of wickets by 1 as well, I made an identical monostable for the 1 wicket button which increments the number shown on the wickets 7-segment display by 1 when triggered. As the runs score passes from 009-010, the Q10 (÷10) output on the 4026 responsible for the 1s digit sends a pulse to the clock input of the 4026 responsible for the 10s digit which causes the 7-segment display showing the 10s digit to increment by 1 . In the same way, when the score passes from 099 to 100, the Q10 output of the 10s digit 4026 pulses high which causes the 100s digit 7-segment display to increment by 1. This continues until the number of wickets displayed changes from 9 to 0 or the number of overs bowled has finished. This means that the team batting have finished scoring their runs and it's then time for the next team to bat. The score of the first team is then recorded and the reset switch is pressed again which causes the score to go back to 000-0. Then, the second team's runs and wickets are scored in the same way that the first team's was done. This time, the scoring and match stops when either the second team passes the score of the first team before the overs have finished or the number of wickets lost by the second team passes from 9 to 0. While this is happening, the light sensing subsystem, consisting of an LDR, resistors and an op-amp, is constantly measuring the light level using an LDR. This is because the LDR's resistivity increases as light level decreases so as the light level falls below 425 lux, the voltage across the resistor below the LDR decreases and passes the threshold voltage of 4.48V. And as this happens, the op-amp outputs a high signal to the row of white LEDs, essentially converting the analogue input from the light sensing subsystem into a binary/digital output for the LEDs. Therefore, as all of the above is in accordance with my specification (including voltages across, current through and power dissipated by the necessary components), the entire system works as expected.

Signal Transfer

In terms of signal transfer, I feel there are 3 key parts that incorporate signal transfer. The first is the logic needed to AND the output of the astable and the output of the monostable triggered. This is because the logic outputs 1 when the output of the astable is high and the output of the monostable is high. Since the output of both the astable and monostable are square waves, this means that an ADC isn't necessary and the output of the logic can go into the clock input of the 4026. For more information on how the logic itself works, please see page 16. Another point is where the runs score goes from 001 to 010. This is because the output of the divide by 10 output of the ones digit 4026 goes into the clock input of the tens digit 4026. This ensures that the digital output of the divide by 10 pin causes the value on the 7 segment display to increment by 1 as the clock input is rising edge triggered. The last part involving signal transfer is the light sensing subsystem which converts the analogue output of the LDR into a digital output for the LEDs to light up at a specific light intensity. This is done with the help of an op-amp. Since the arrangement of the op-amp was such that it worked as a comparator, this meant that when the non-inverting input is greater than the inverting input, the op-amp would be positively saturated, outputting 5V during testing. I used this feature of the comparator to ensure that the voltage going to the row of LEDs was 5V.

Improvements

1. A switch can be connected in series with the power supply and then to the circuit so that the switch would act as a means of turning the scoreboard off and on without manually disconnecting and reconnecting the power supply.
2. More buttons can be added to account for 2 runs or 3 runs scored. This would be done by adding 2 more monostables, one which has a time period covering 2 rising edges and the other having a time period covering 3 rising edges. In the same way, 2 more buttons for a wide and no ball can be added as they would serve the same purpose as the 1 run button - by adding 1 to the runs scoreboard.
3. A means of keeping track of the number of balls bowled could be implemented. This would be done with a 4510 counter and a 4511 decoder as the counter would be triggered every time a ball was bowled and when it reaches 6, which is the number of balls in an over, the counter would reset back to 0 for the next over and doing so would cause another two 4026s connected to two more 7-segment displays (one for the ones digit and the other for the 10s digit) to increase the value on the 7-segments by 1, showing the end of an over.
4. Another 4026 and 7-segment display can be added for the wickets scorecard so that it can show a 2 digit value for the number of wickets lost. However, since this would mean that the wickets score could go up to 19, this may not be the most ideal solution.
5. Instead of resetting the scoreboard in between innings, another 3 4026s and 3 7-segment displays could be added to continue displaying the number of runs scored by the previous team so that anyone watching knows how many runs the second team needs to win.
6. Lastly, for the above improvements to take place, the reset switch would have to connect to every 4026 which has been added so that all the 7-segment displays show 0.